

微波电路原理与设计

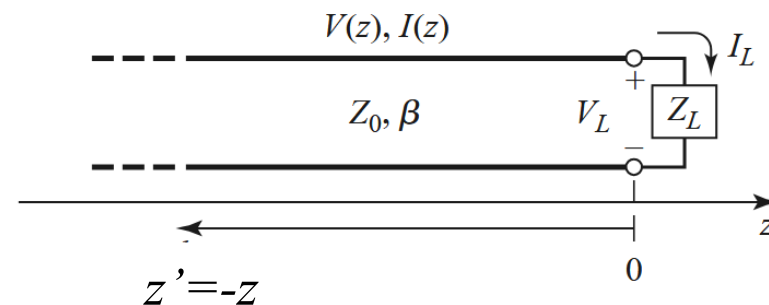
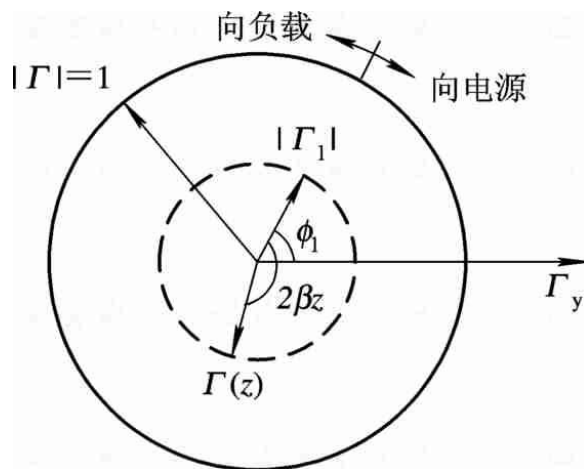
第一章 史密斯圆图及阻抗匹配

张允晶
2026-03

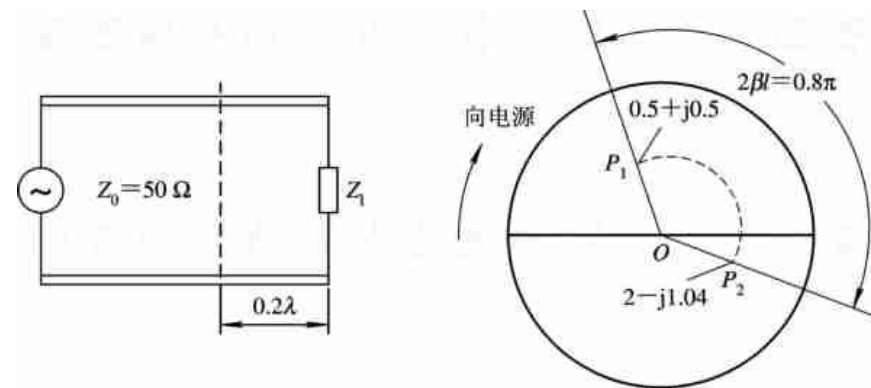
史密斯阻抗圆图

$$\Gamma(z) = \frac{Z_{in}(z') - Z_0}{Z_{in}(z') + Z_0} = \frac{\bar{Z}_{in}(z') - 1}{\bar{Z}_{in}(z') + 1}$$

$$\Gamma(z') = \frac{V_0^- e^{-j\beta z'}}{V_0^+ e^{j\beta z'}} = \Gamma_0 e^{-j2\beta z'}$$



[例] 已知传输线的特性阻抗 $Z_0=50\Omega$ ，如图1-22所示。假设传输线的负载阻抗为 $Z_1=25+j25\Omega$ ，求离负载 $z=0.2\lambda$ 处的等效阻抗。

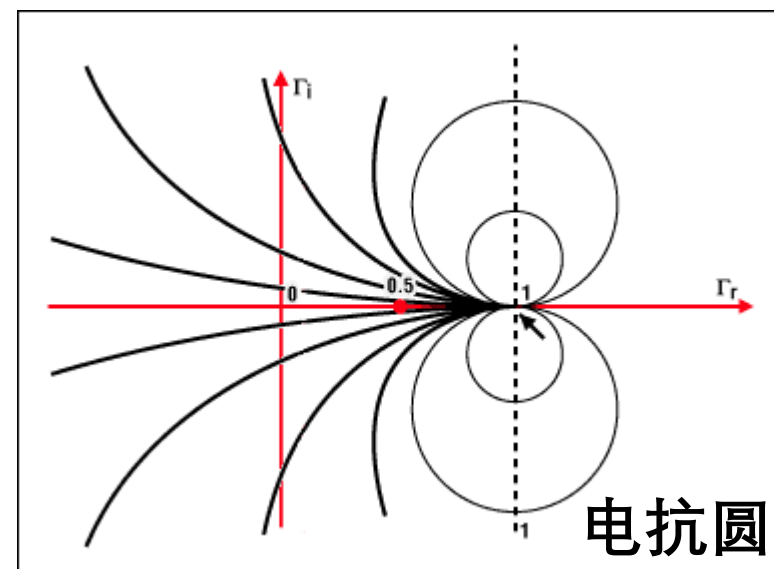
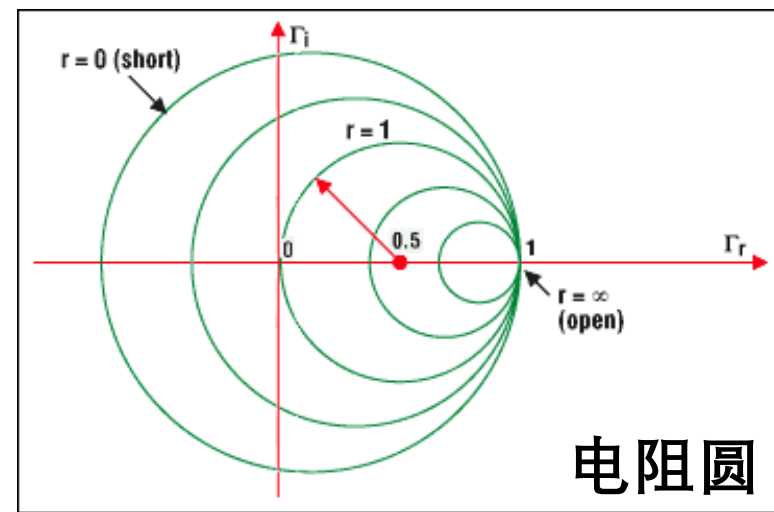
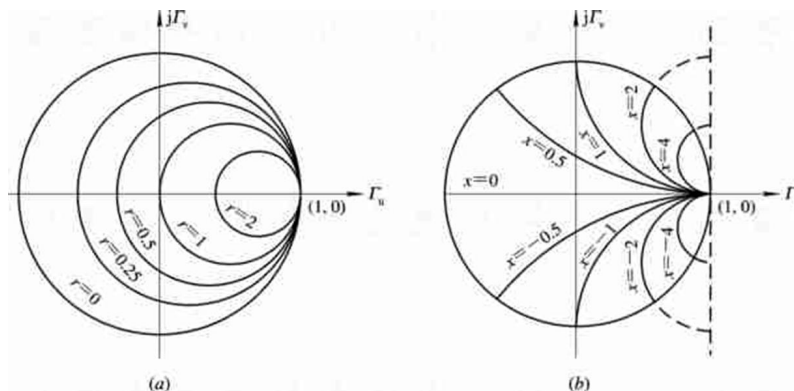


史密斯阻抗圆图

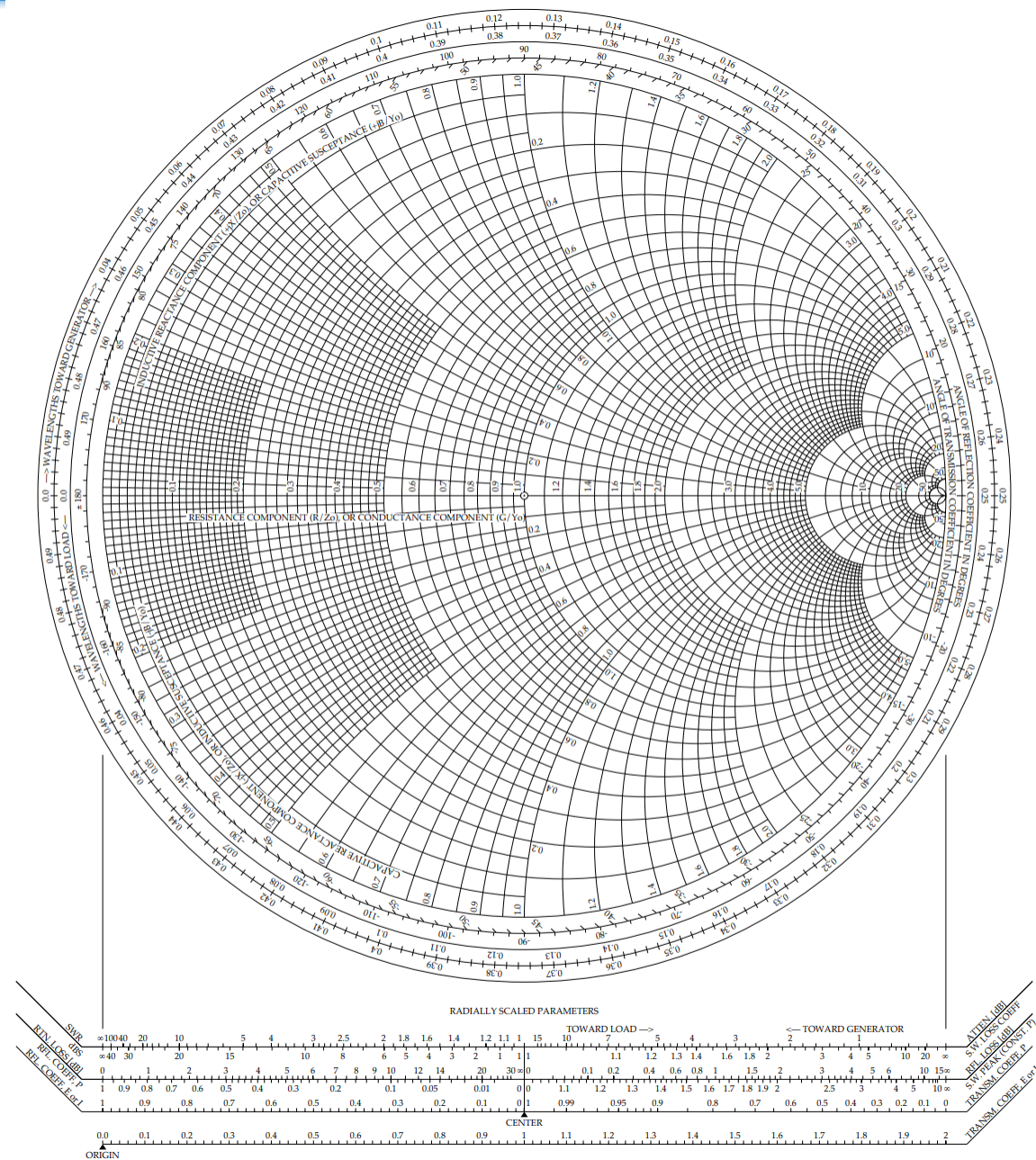
$$\Gamma(z) = \Gamma_r + j\Gamma_i \quad \bar{Z}_{in} = \frac{Z_{in}}{Z_0} = \frac{1 + (\Gamma_r + j\Gamma_i)}{1 - (\Gamma_r + j\Gamma_i)}$$

令 $\bar{Z}_{in} = r + jx$, 则可得以下方程:

$$\left\{ \begin{aligned} \left(\Gamma_r - \frac{r}{1+r} \right)^2 + \Gamma_i^2 &= \left(\frac{1}{1+r} \right)^2 \\ (\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x} \right)^2 &= \left(\frac{1}{x} \right)^2 \end{aligned} \right.$$



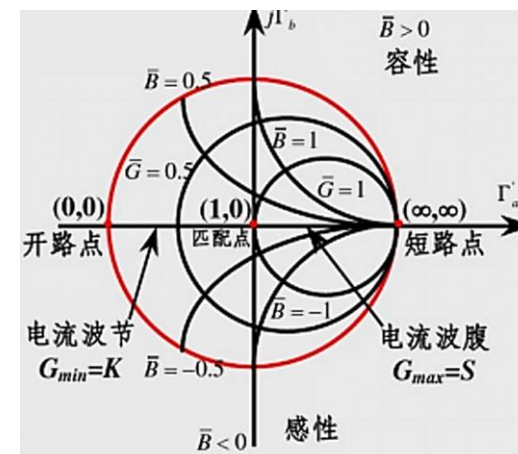
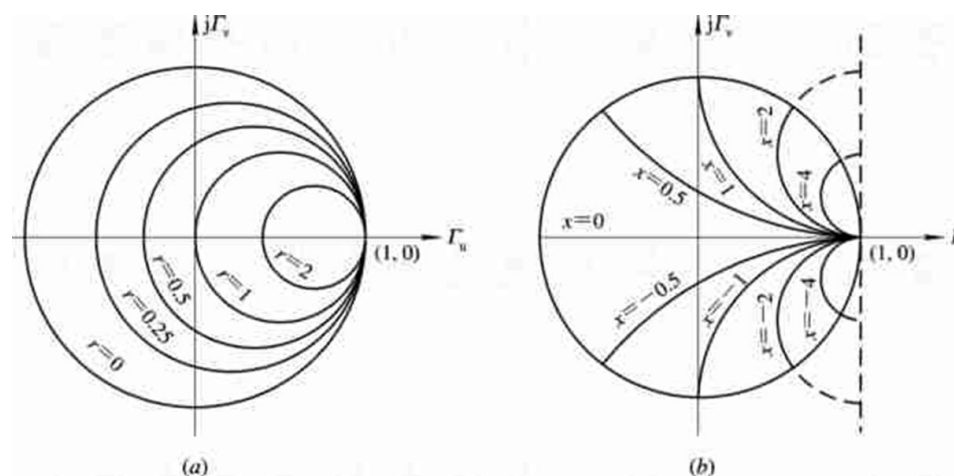
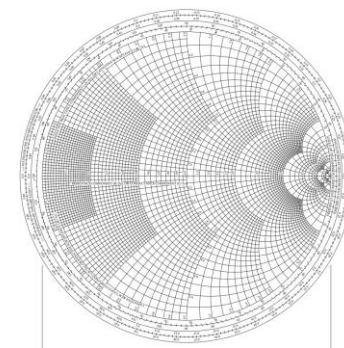
史密斯阻抗圆图



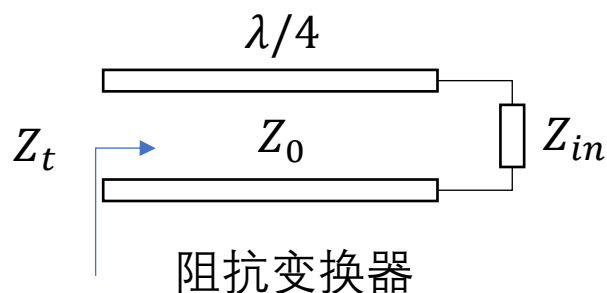
史密斯阻抗圆图

$$\rho = \text{SWR} = \frac{V_{\max}}{V_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

1. 阻抗圆图上半圆内的电抗 $x > 0$ 呈感性，下半圆内的电抗 $x < 0$ 呈容性。
2. 实轴上的点代表纯电阻点，左半轴上的点为电压波节点，代表行波系数 k (Why?)
3. 右半轴上的点为电压波腹点，其上的刻度代表驻波比 ρ (Why?)
4. 圆图旋转一周为 $\lambda/2$
5. $|\Gamma| = 1$ 的圆周上的点代表纯电抗点
6. 实轴左端点为短路点，右端点为开路点，中心点是匹配点
7. 在传输线上由负载向电源方向移动时，在圆图上应顺时针旋转；反之，由电源向负载方向移动时，应逆时针旋转。

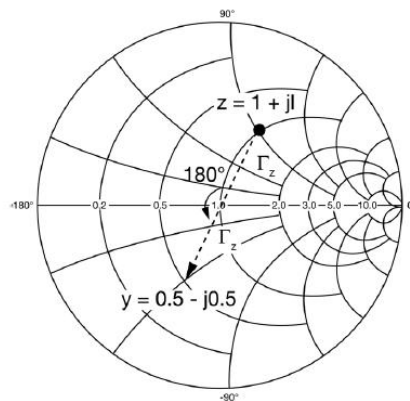


史密斯导纳圆图

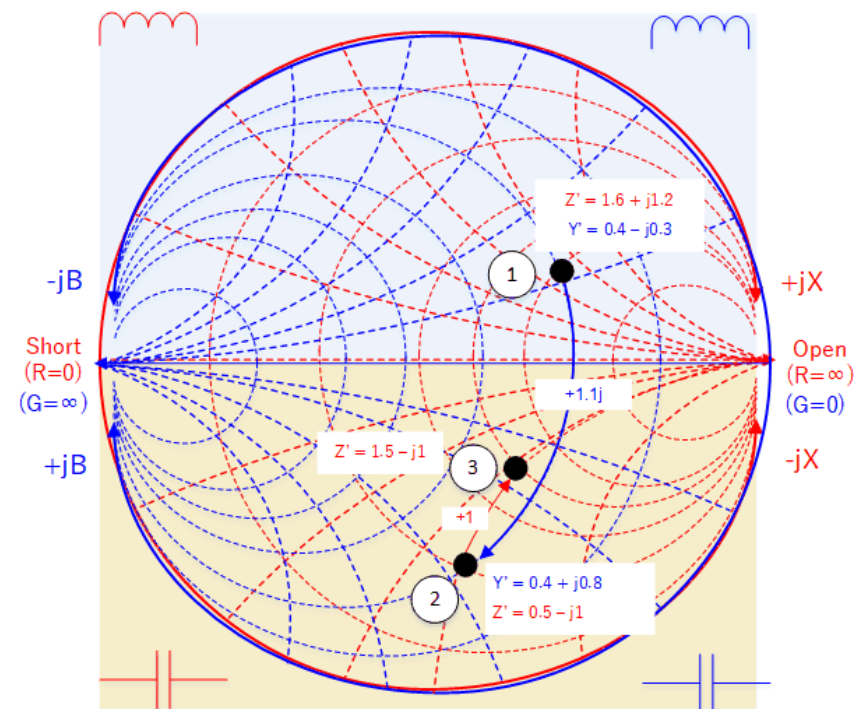


$$Z_t = Z_0^2 / Z_{in}$$

$$z_t = \frac{Z_t}{Z_0} = \frac{Z_0}{Z_{in}} = \frac{1}{z_{in}} = y_{in} = g + jb$$



$$\Gamma_z = \frac{z-1}{z+1} \quad \Gamma_y = \frac{1-y}{1+y} \quad |y| = \frac{1}{|z|}$$



- ◆ 经过1/4波长，原来的归一化阻抗变成了归一化导纳，
在阻抗圆图直接读出的数值即为转化后的导纳值
- ◆ Z转化为Y后，其反射系数的幅值不变，相位发生180度变化

史密斯圆图的应用-串联RL

$$z_{in}(\omega_L) = r + j \frac{\omega_L L}{Z_0}$$

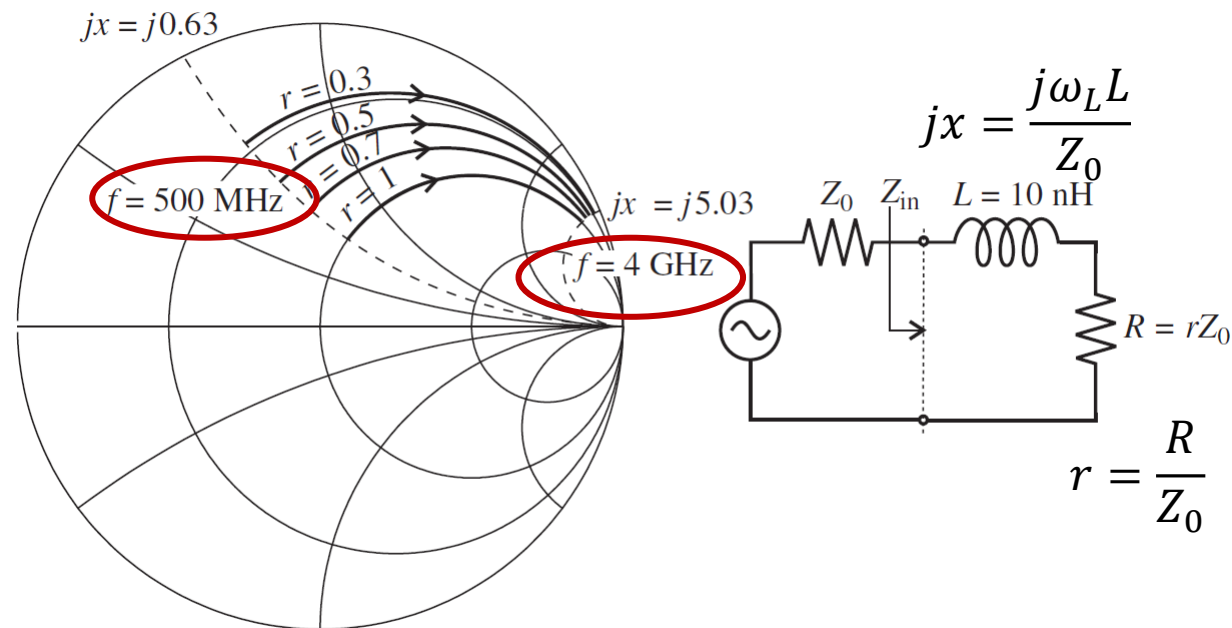
- 在等电阻圆上走，相当于串联电阻不变，增加或减少一定量的电抗(变化的可以是 ω_L 或者是 L)
- 在等电抗圆上走，相当于串联电抗不变，增加或减少一定量的电阻 (变化的为 r)

$$jx(500 \text{ MHz}) = j \frac{2\pi \times 500e6 \times 10e-9}{50} = j0.63$$

增加频率 ↓ 等电阻圆上从j0.63→j5.03

$$jx(4000 \text{ MHz}) = j \frac{2\pi \times 4000e6 \times 10e-9}{50} = j5.03$$

阻抗图



史密斯圆图的应用-串联RL

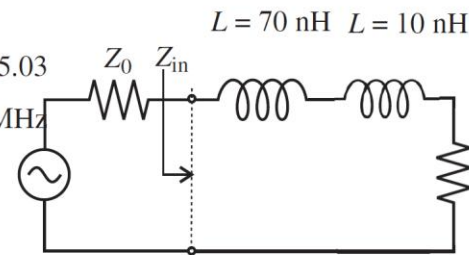
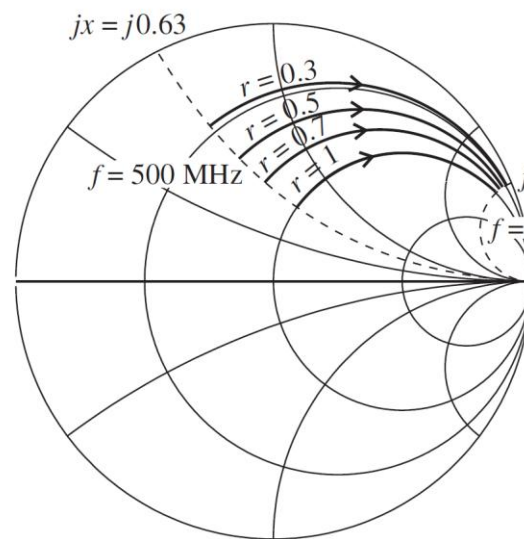
串联（增加）电感

$$z_{in}(\omega_L) = r + jx + j \frac{\omega_L \Delta L}{Z_0}$$

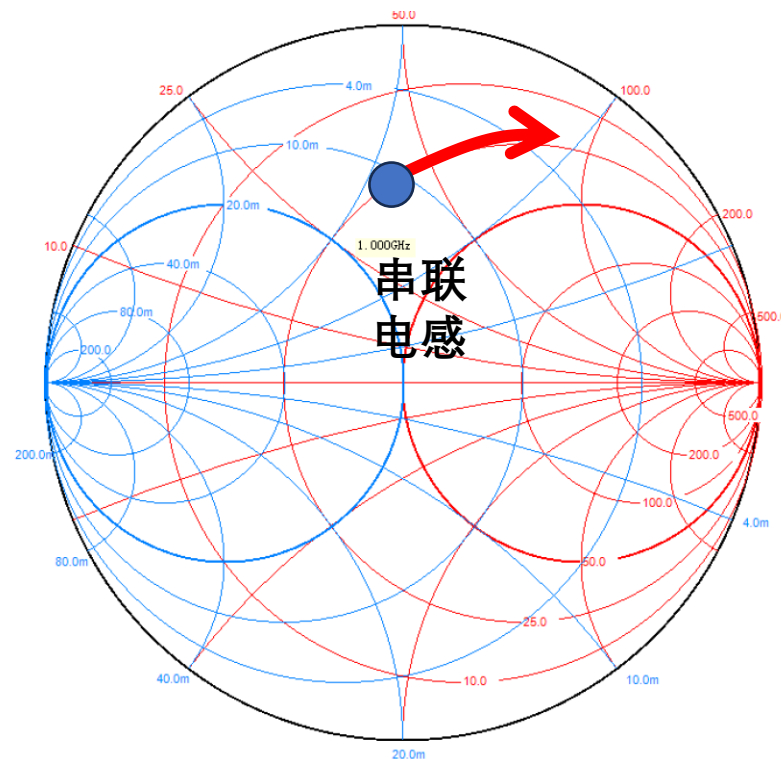
$$jx(10 \text{ nH}) = j \frac{2\pi \times 500e6 \times 10e-9}{50} = j0.63$$

↓ 等电阻圆上从j0.63→j5.03

$$\begin{aligned} jx(10 \text{ nH} + 70 \text{ nH}) &= j0.63 + j \frac{2\pi \times 500e6 \times 70e-9}{50} \\ &= j0.63 + j4.4 = j5.03 \end{aligned}$$



串联电感，沿电阻圆顺时针走



史密斯圆图的应用-串联RC

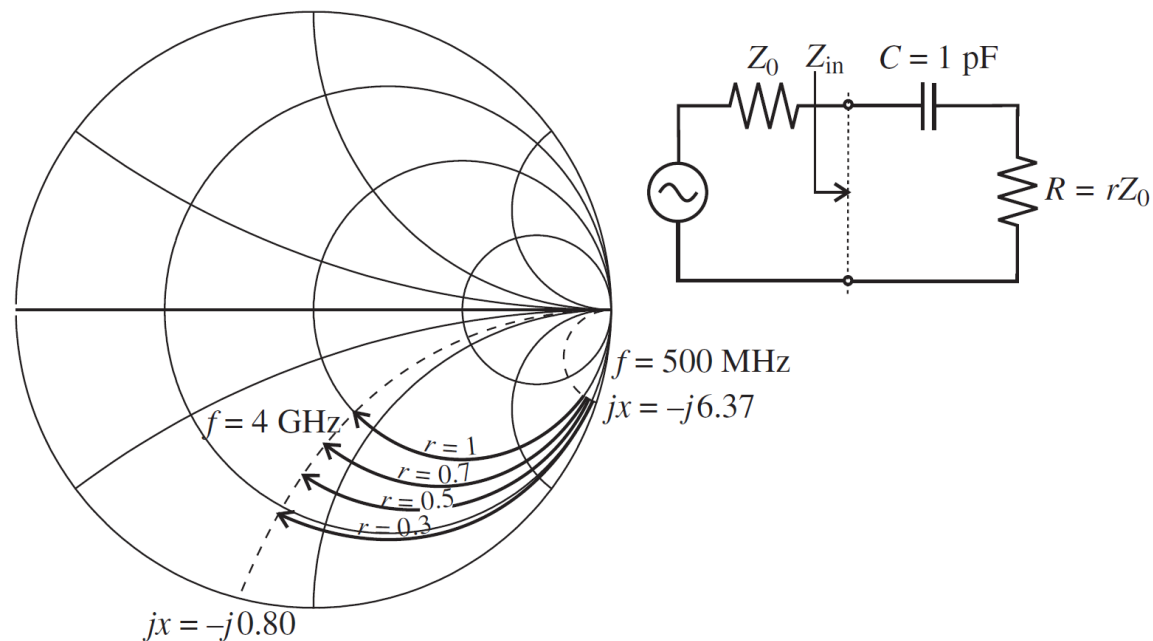
$$z_{in}(\omega_L) = r - j \frac{1}{\omega C Z_0}$$

- 在等电阻圆上走，相当于串联电阻不变，增加或减少一定量的电抗(变化的可以是 ω_L 或者是 C)
- 在等电抗圆上走，相当于串联电抗不变，增加或减少一定量的电阻 (变化的为 r)

$$jx(500 \text{ MHz}) = -j \frac{1}{\omega_0 C Z_0} = -j6.37$$

增加频率 ↓

$$jx(4000 \text{ MHz}) = -j \frac{1}{\omega_1 C Z_0} = -j0.8$$

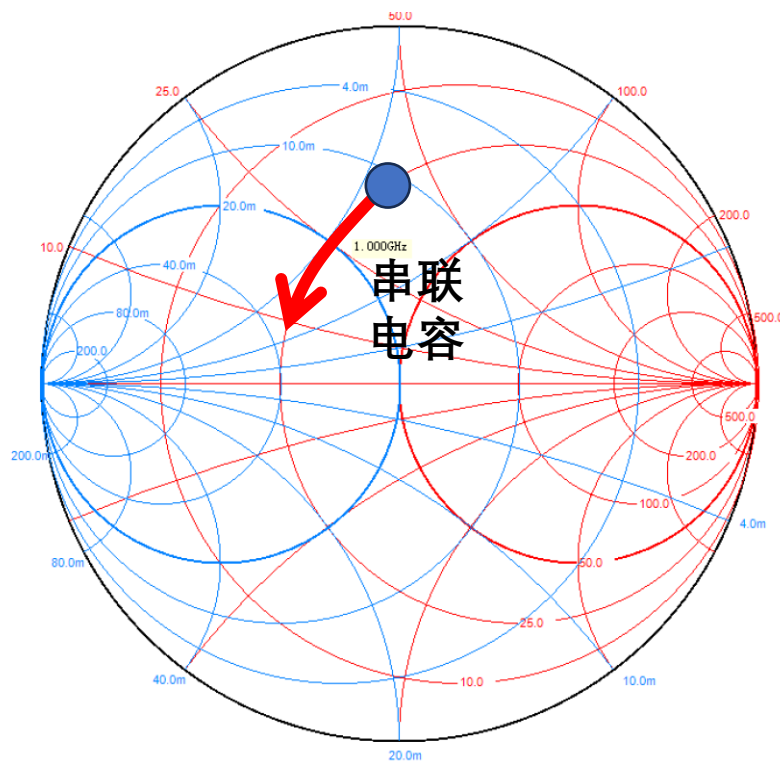


史密斯圆图的应用-串联RC

串联（增加）电容

$$z_{in}(\omega_L) = r + jx - j \frac{1}{\omega_L \Delta C Z_0}$$

串联电容，沿电阻圆逆时针走

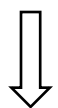


史密斯圆图的应用-并联RL

$$y_{in}(\omega_L) = g - j \frac{Z_0}{\omega_L L}$$

- 在**等电导圆**上走，相当于并联电阻不变，增加或减少一定量的电抗(变化的可以是 ω_L 或者是 L)
- 在**等电纳圆**上走，相当于并联电抗不变，增加或减少一定量的电导（变化的为 g ）

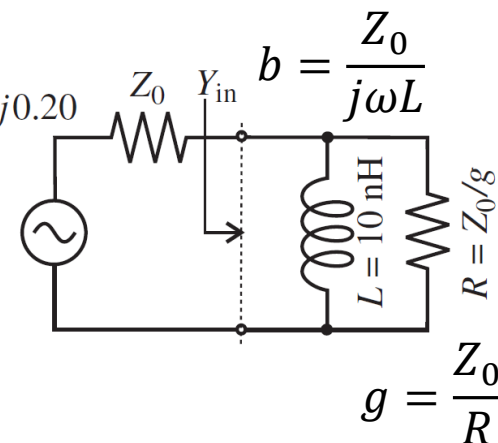
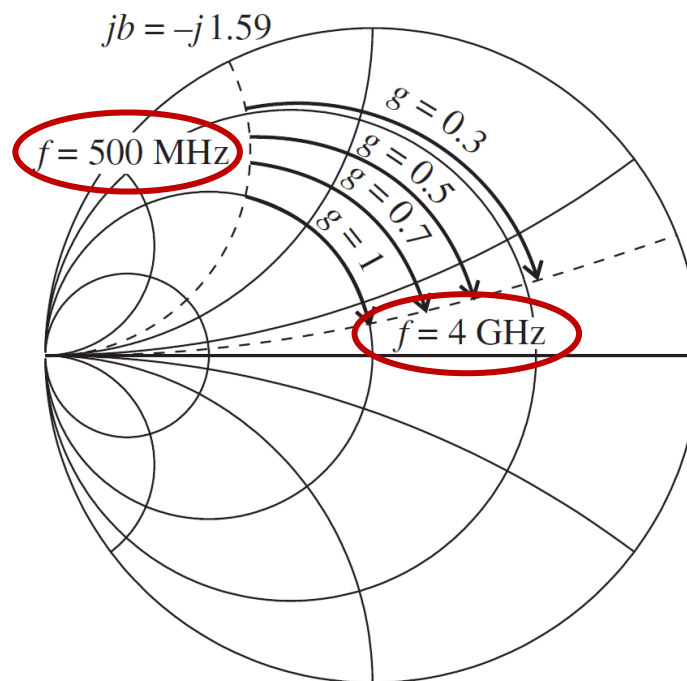
$$jb(500 \text{ MHz}) = -\frac{j50}{2\pi \times 500e6 \times 10e-9} = -j1.59$$



等电导圆上从-j1.59→-j0.2

$$jb(4000 \text{ MHz}) = -\frac{j50}{2\pi \times 4000e6 \times 10e-9} = -j0.2$$

导纳图

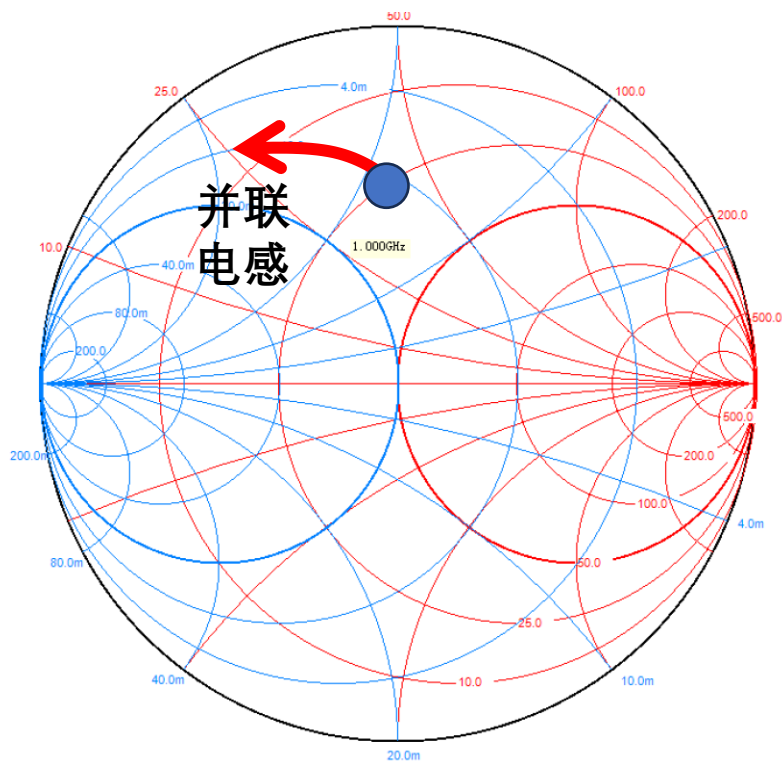


史密斯圆图的应用-并联RL

并联电感

$$z_{in}(\omega_L) = g + jb - j \frac{Z_0}{\omega_L \Delta L}$$

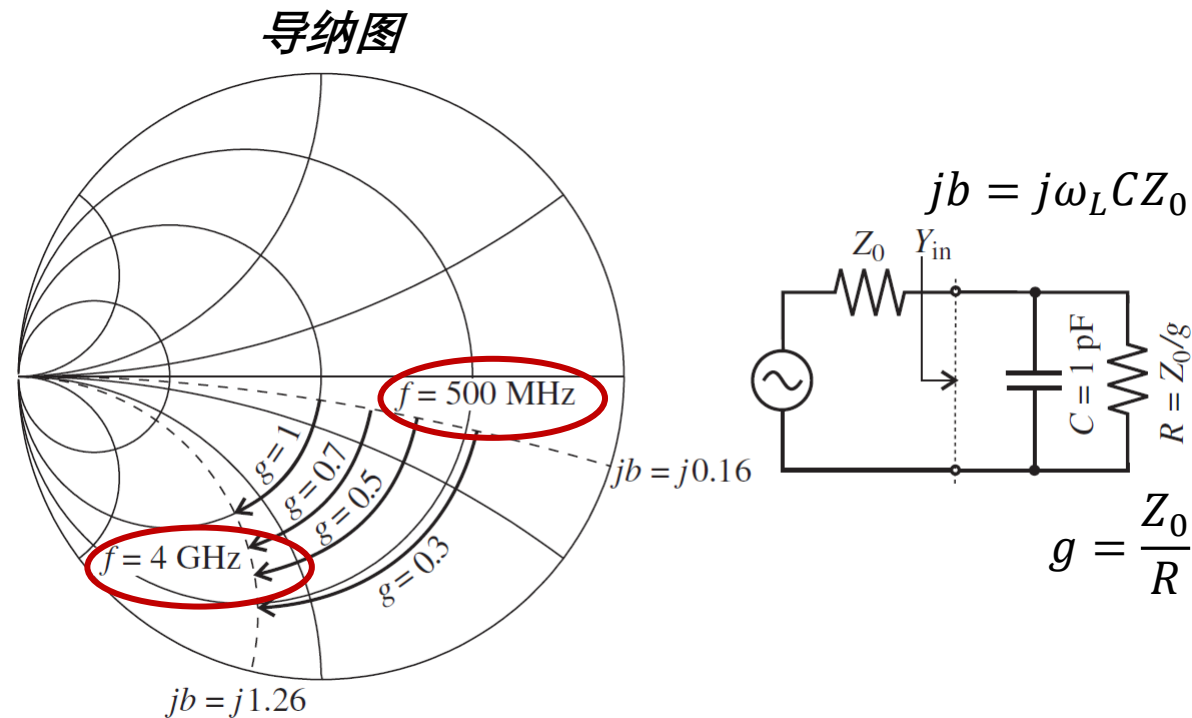
并联电感，沿电导圆逆时针走



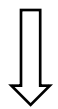
史密斯圆图的应用-并联RC

$$y_{in}(\omega_L) = g + j\omega_L C Z_0$$

- 在**等电导圆**上走，相当于并联电阻不变，增加或减少一定量的电抗(变化的可以是 ω_L 或者是 C)
- 在**等电纳圆**上走，相当于并联电抗不变，增加或减少一定量的电导 (变化的为 g)



$$jb(500 \text{ MHz}) = j2\pi \times 500e6 \times 50 \times 1e-12 = j0.16$$



等电导圆上从j0.16→j1.26

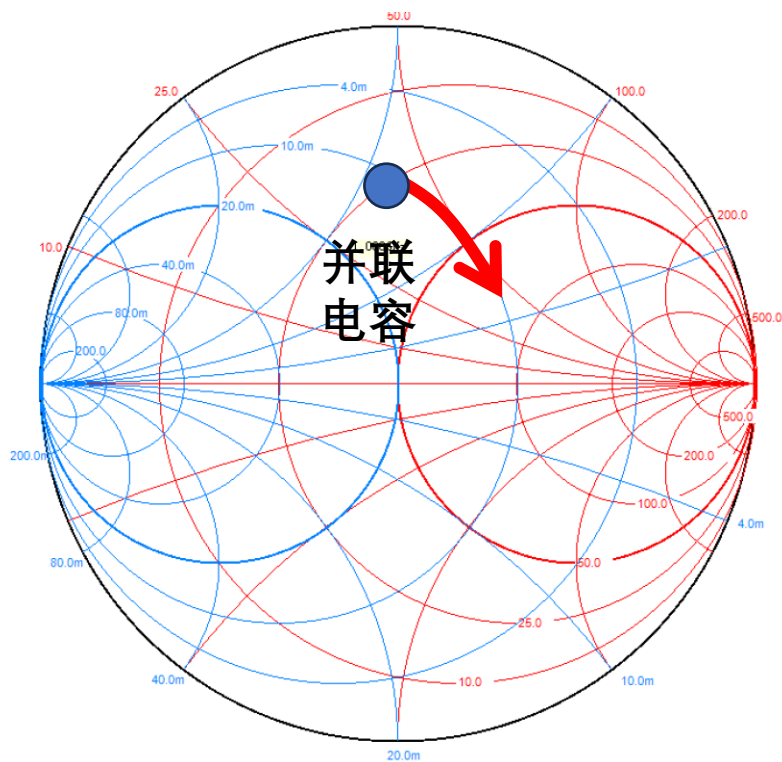
$$jb(4000 \text{ MHz}) = j2\pi \times 4000e6 \times 50 \times 1e-12 = j1.26$$

史密斯圆图的应用-并联RL

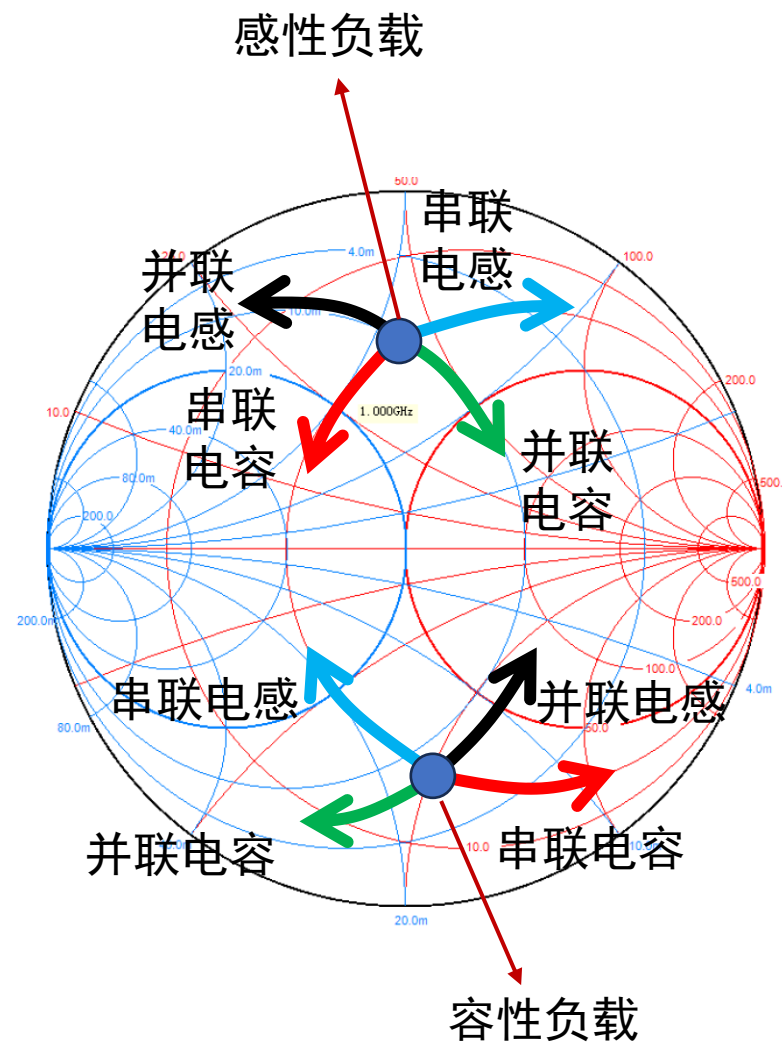
并联电容

$$z_{in}(\omega_L) = g + jb + j\omega\Delta CZ_0$$

并联电容，沿电导圆顺时针走

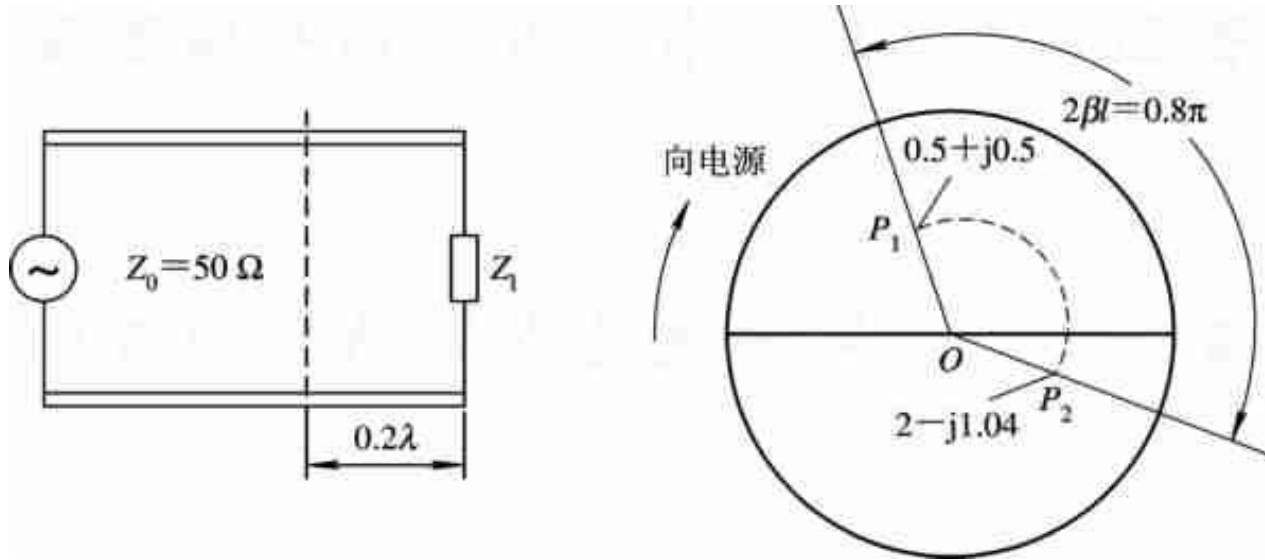


史密斯圆图的应用-串并联电感/电容



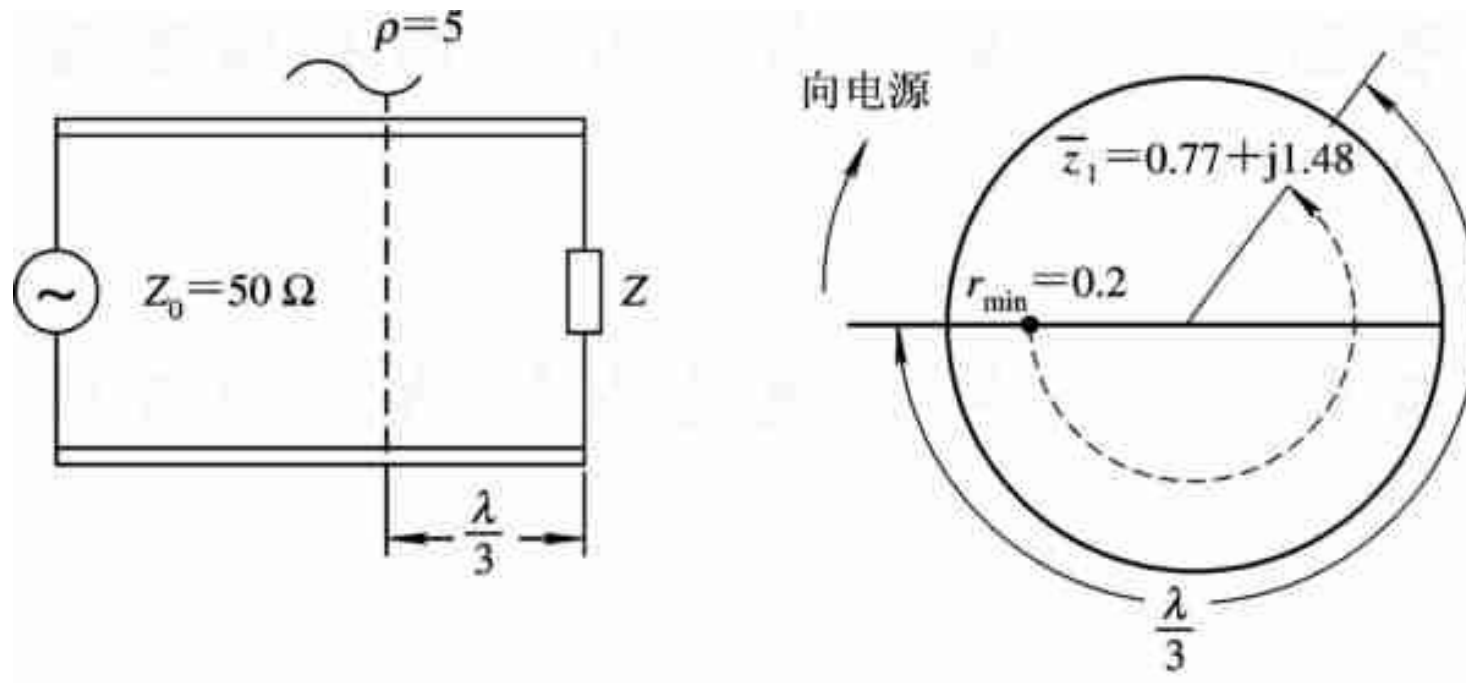
史密斯圆图的应用-传输线的影响

[例] 已知传输线的特性阻抗 $Z_0=50\Omega$ ，如图1-22所示。假设传输线的负载阻抗为 $Z_L=25+j25\Omega$ ，求离负载 $z=0.2\lambda$ 处的等效阻抗。

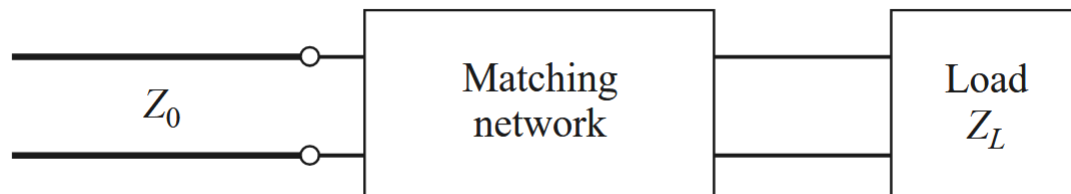


史密斯圆图的应用-传输线的影响

[例] 在特性阻抗 $Z_0=50\Omega$ 的无耗传输线上测得驻波比 $\rho=5$ ，电压最小点出现在 $z=\lambda/3$ 处，求负载阻抗。

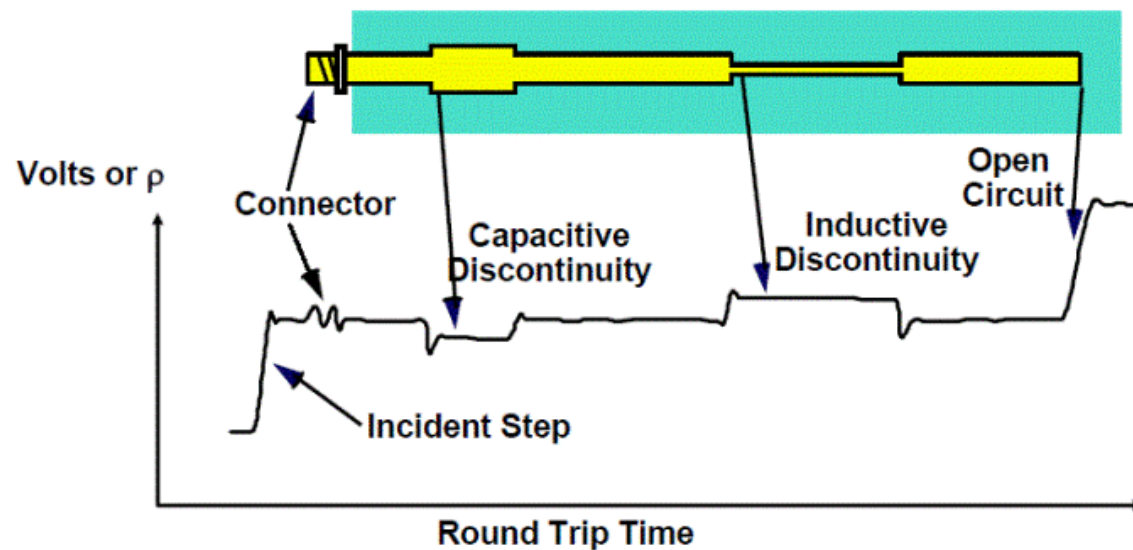


阻抗匹配

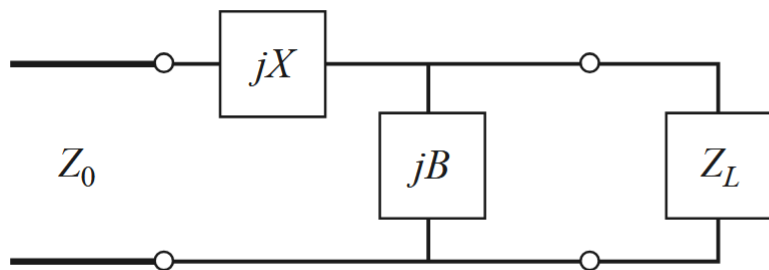


阻抗匹配考虑原则:

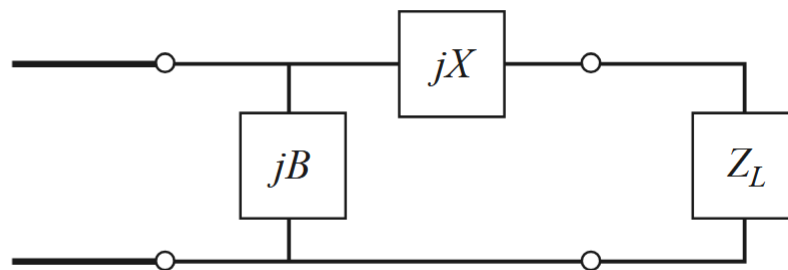
1. 一般情况下，要求匹配网络无损耗元件，且整体损耗低
2. 结构是否简单？
3. 带宽如何？
4. 实施难度如何？
5. 是否可调？



集总元件阻抗匹配



(a)



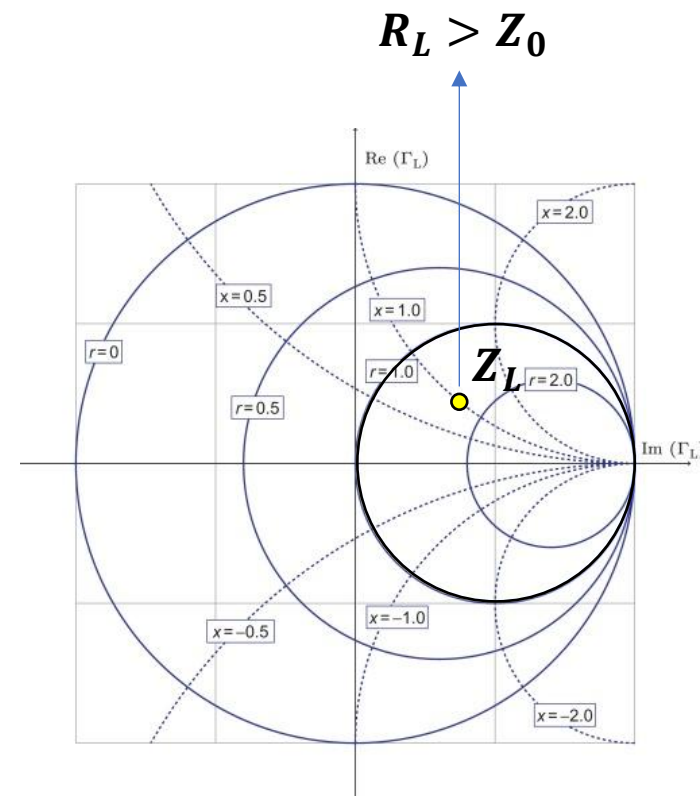
(b)

$$Z_0 = jX + \frac{1}{jB + 1/(R_L + jX_L)} \Rightarrow$$

$$\begin{aligned} B(XR_L - X_L Z_0) &= R_L - Z_0, \\ X(1 - BX_L) &= BZ_0 R_L - X_L. \end{aligned}$$

$$\Rightarrow B = \frac{X_L \pm \sqrt{R_L/Z_0} \sqrt{R_L^2 + X_L^2 - Z_0 R_L}}{R_L^2 + X_L^2}.$$

$$\xRightarrow{R_L > Z_0} X = \frac{1}{B} + \frac{X_L Z_0}{R_L} - \frac{Z_0}{B R_L} \xRightarrow{\quad}$$

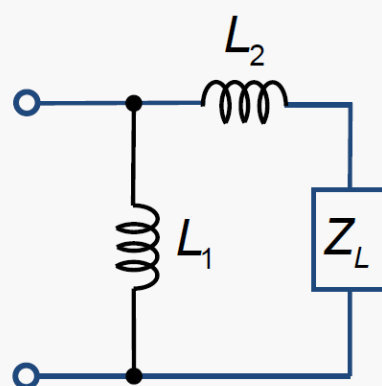
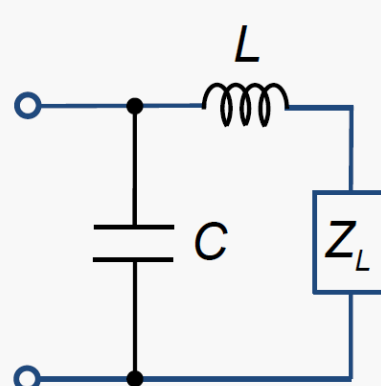
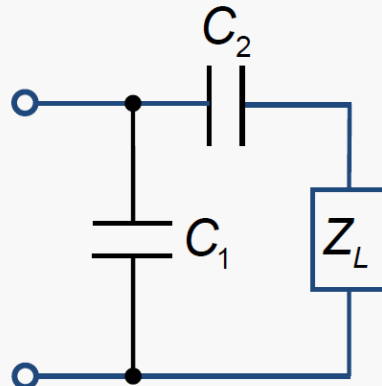
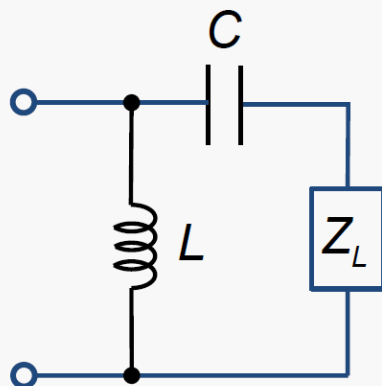
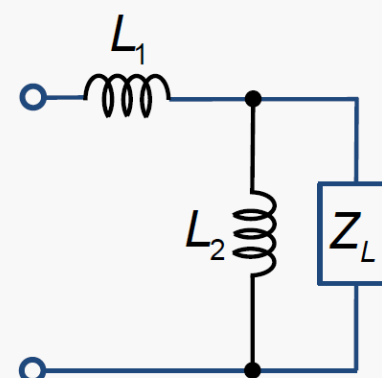
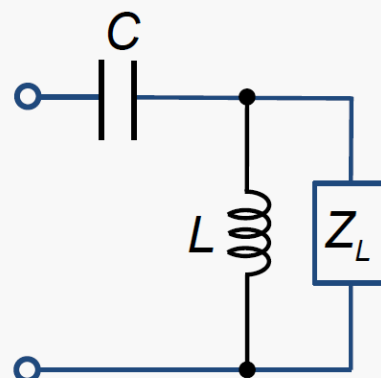
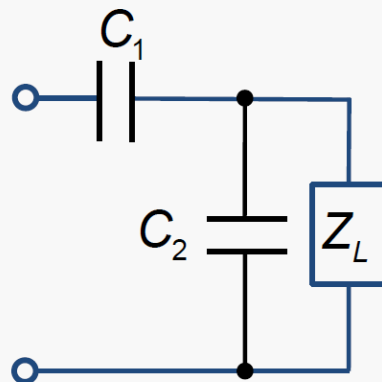
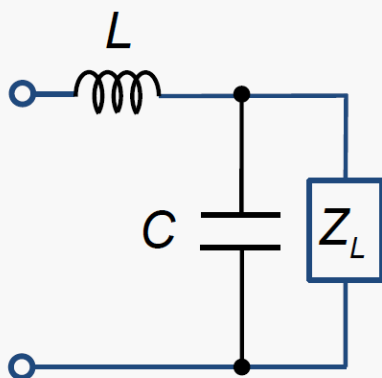


存在两种匹配方式:

B可以为正或为负 (电容或者电感)

X可以为正或为负 (电感或者电容)

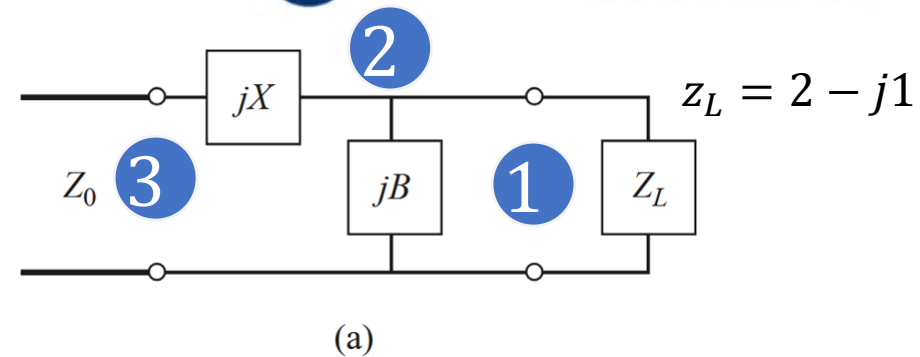
集总元件阻抗匹配-L型匹配



集总元件阻抗匹配

[例] 负载 $Z_L = 200 - j100 \Omega$, 设计一个 L 型匹配网络, 将负载阻抗匹配到 100Ω 的传输线上, 频率为 500 MHz 。

$$z_L = \frac{Z_L}{100} = 2 - j1$$



第一步：1 → 2： 导纳 $+j0.29$

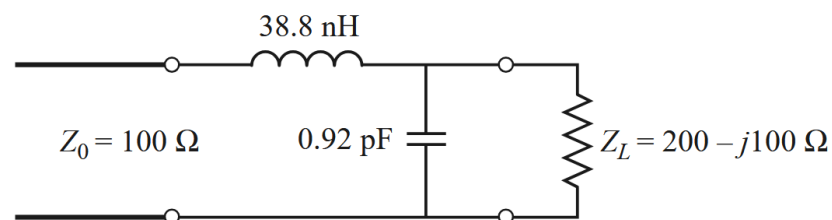
并联电容： $+j\omega C Z_0 = j0.29$

$$C = \frac{0.29}{2\pi \cdot 500e6 \cdot 100} = 0.92 \text{ pF}$$

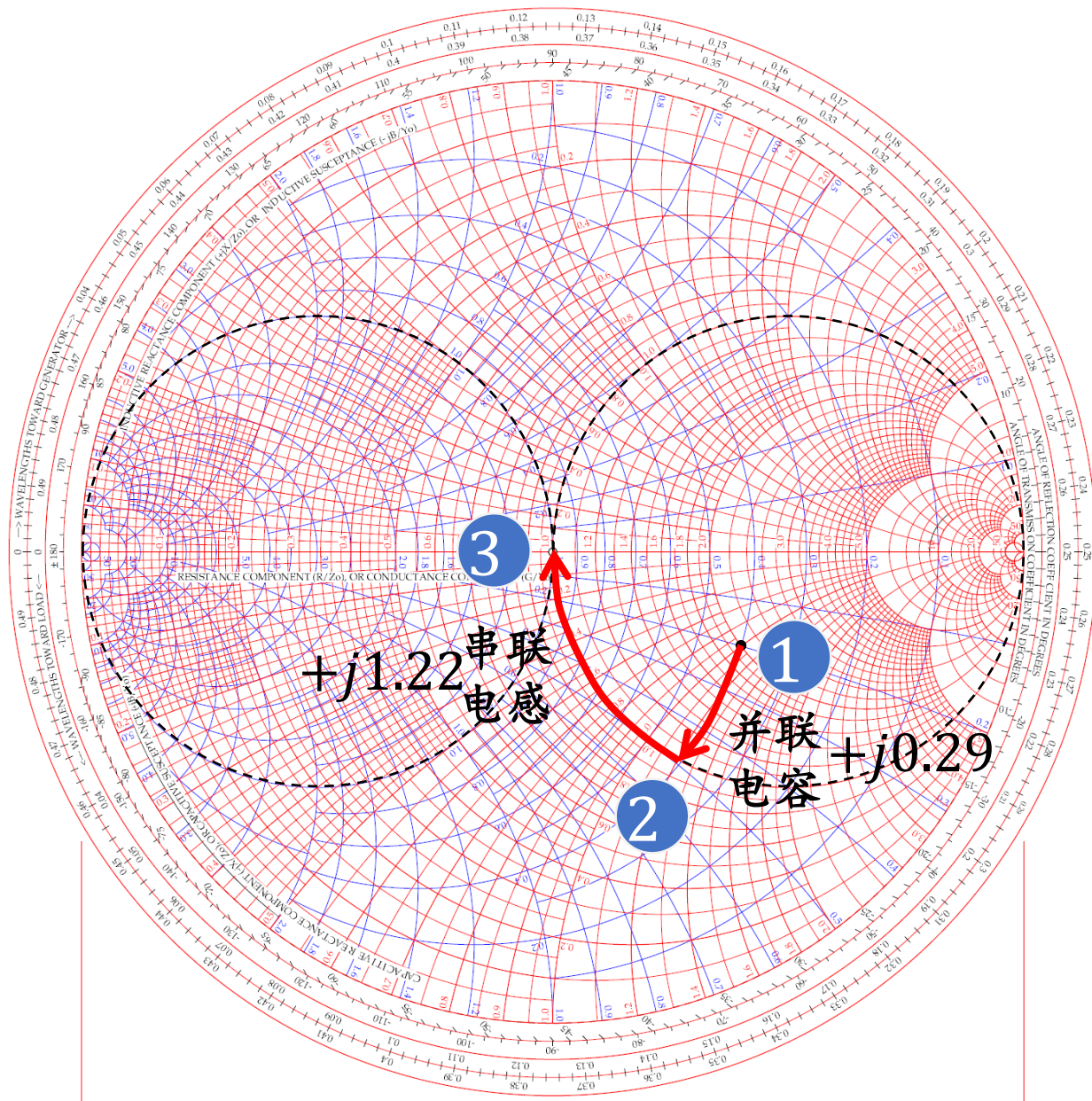
第二步：2 → 3： 阻抗 $+j1.22$

串联电感： $+j\omega L / Z_0 = j1.22$

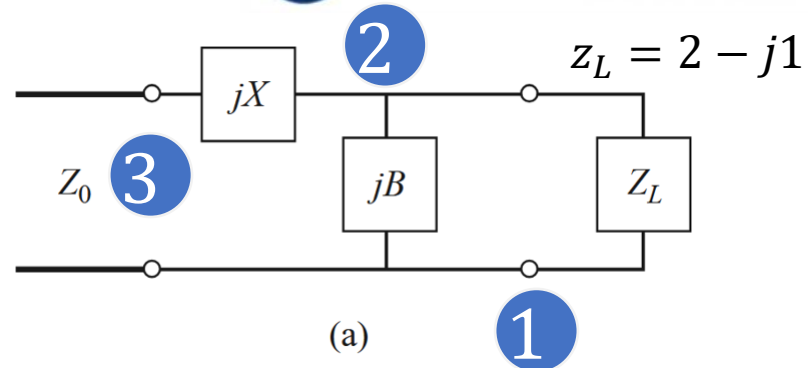
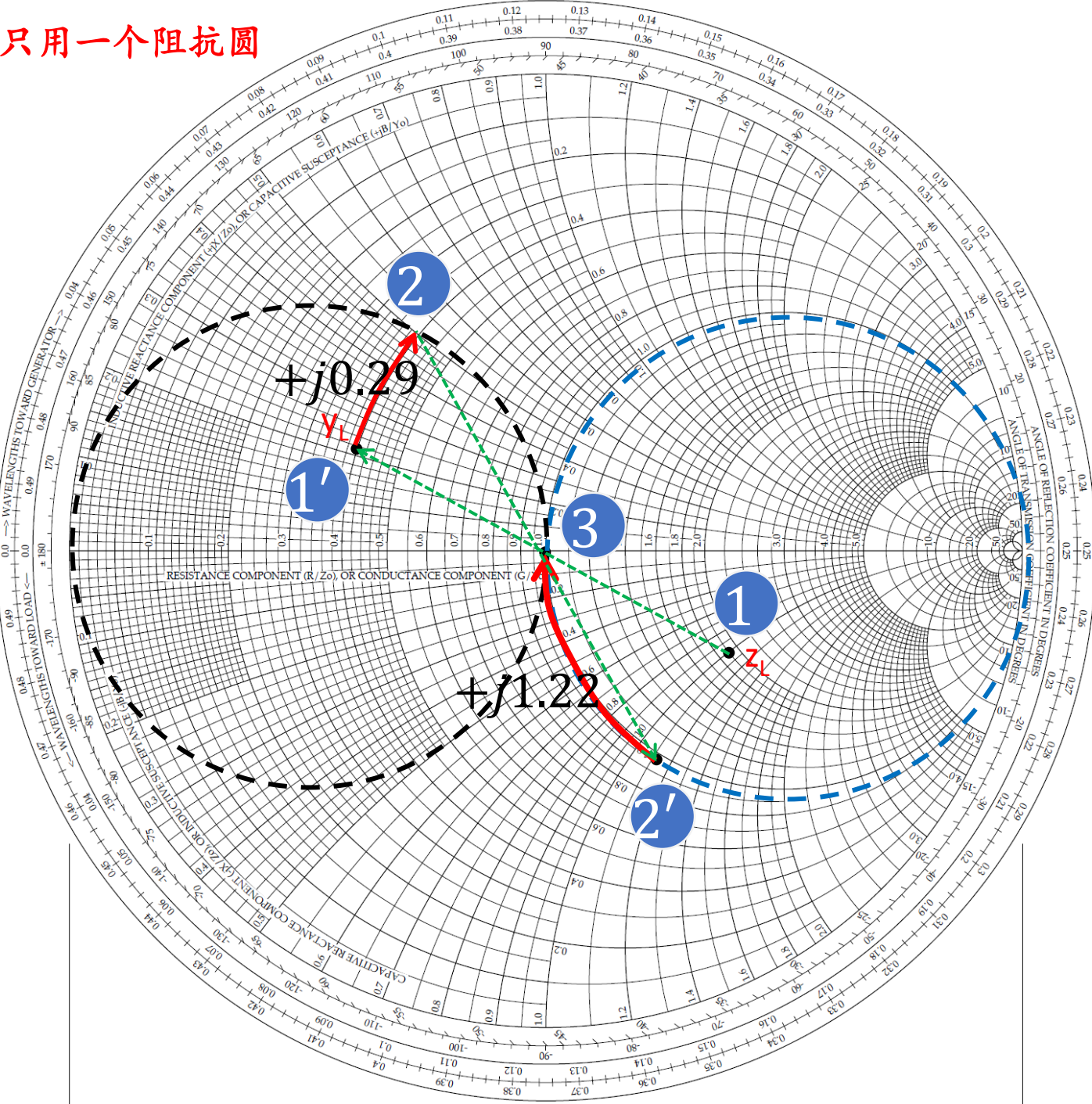
$$L = \frac{1.22 \cdot 100}{2\pi \cdot 500e6} = 38.8 \text{ nH}$$



Solution 1

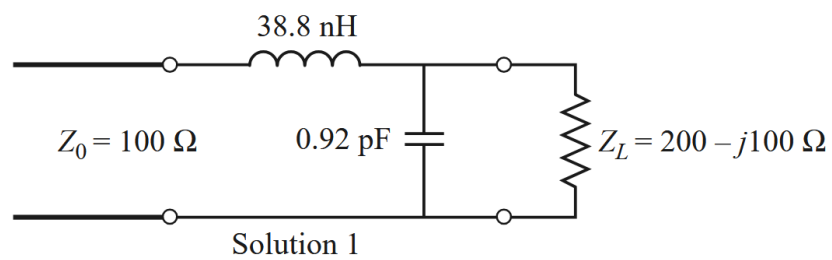


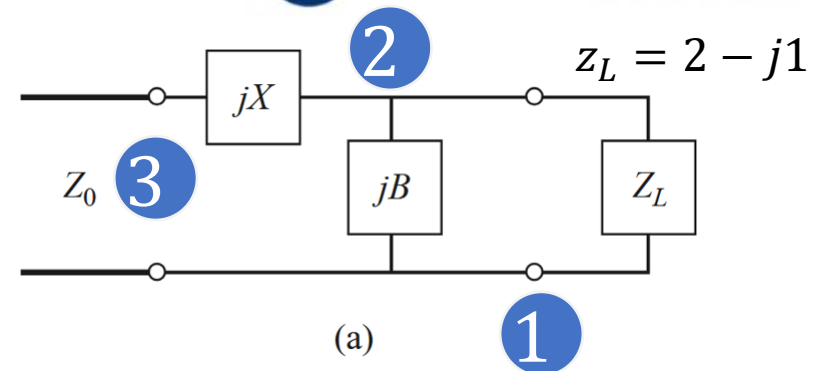
只用一个阻抗圆



第一步：1' → 2: 导纳+j0.29
并联电容： $+j\omega CZ_0 = j0.29$
 $C = \frac{0.29}{2\pi \cdot 500e6 \cdot 100} = 0.92 \text{ pF}$

第二步：2' → 3: 阻抗+j1.22
串联电感： $+j\omega L/Z_0 = j1.22$
 $L = \frac{1.22 \cdot 100}{2\pi \cdot 500e6} = 38.8 \text{ nH}$





第一步：1 → 2: 导纳 $-j0.69$

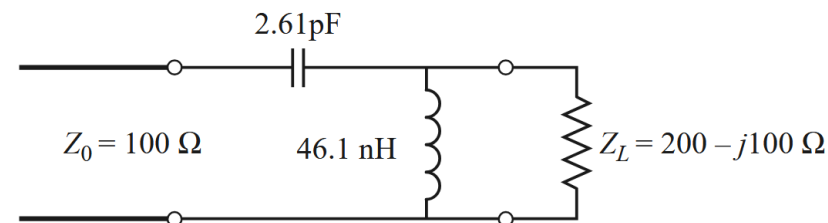
并联电感: $-(jZ_0)/\omega L = -j0.69$

$$L = \frac{100}{0.69 \cdot 2\pi \cdot 500e6} = 46.1 \text{ nH}$$

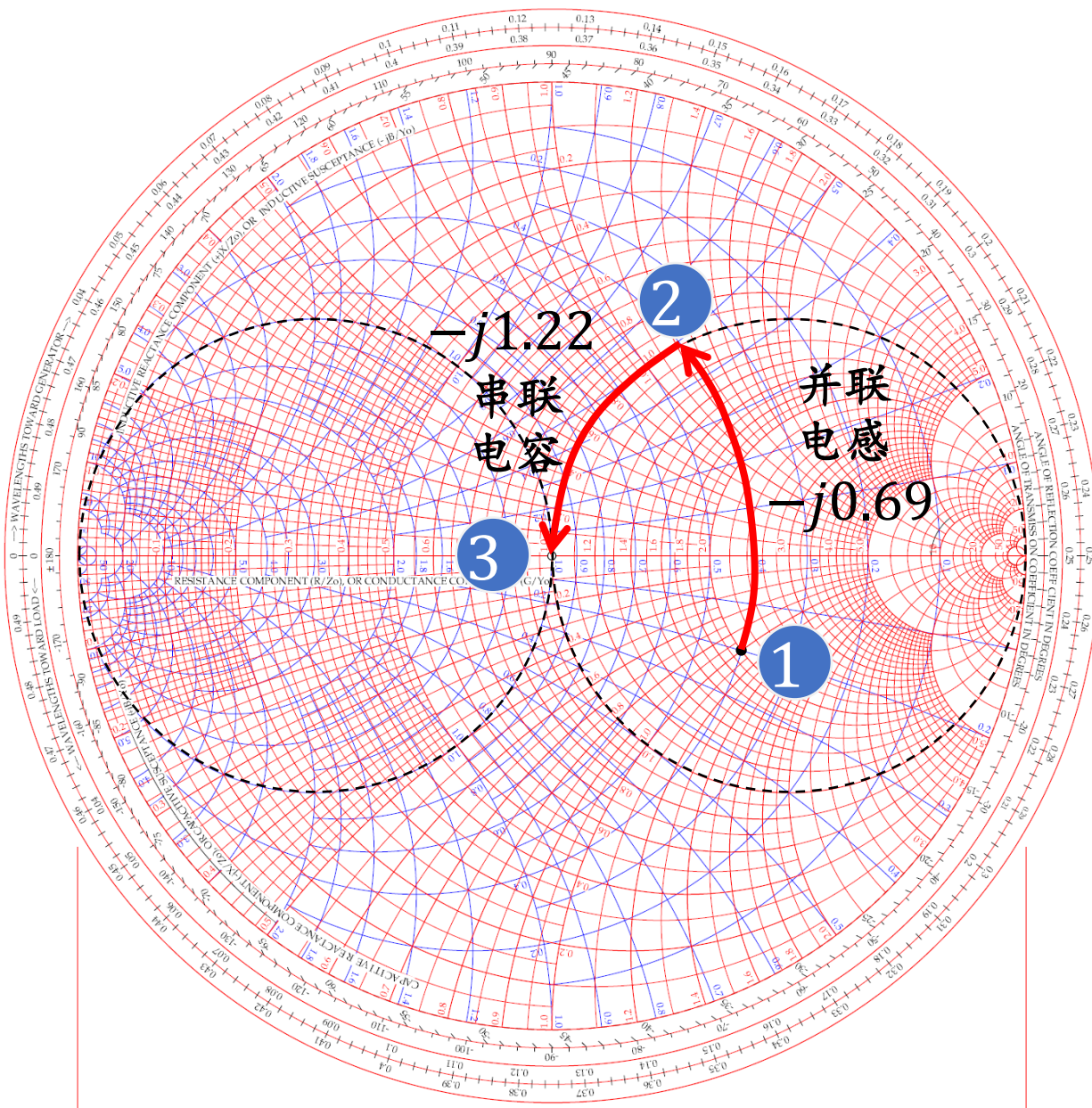
第二步：2 → 3: 阻抗 $-j1.22$

串联电容: $-j1/(\omega CZ_0) = -j1.22$

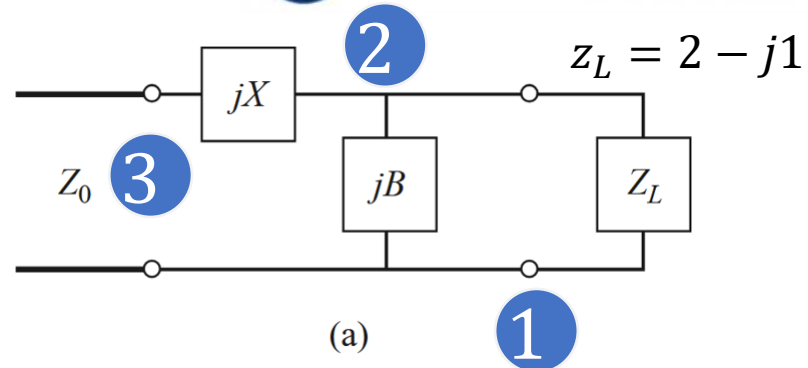
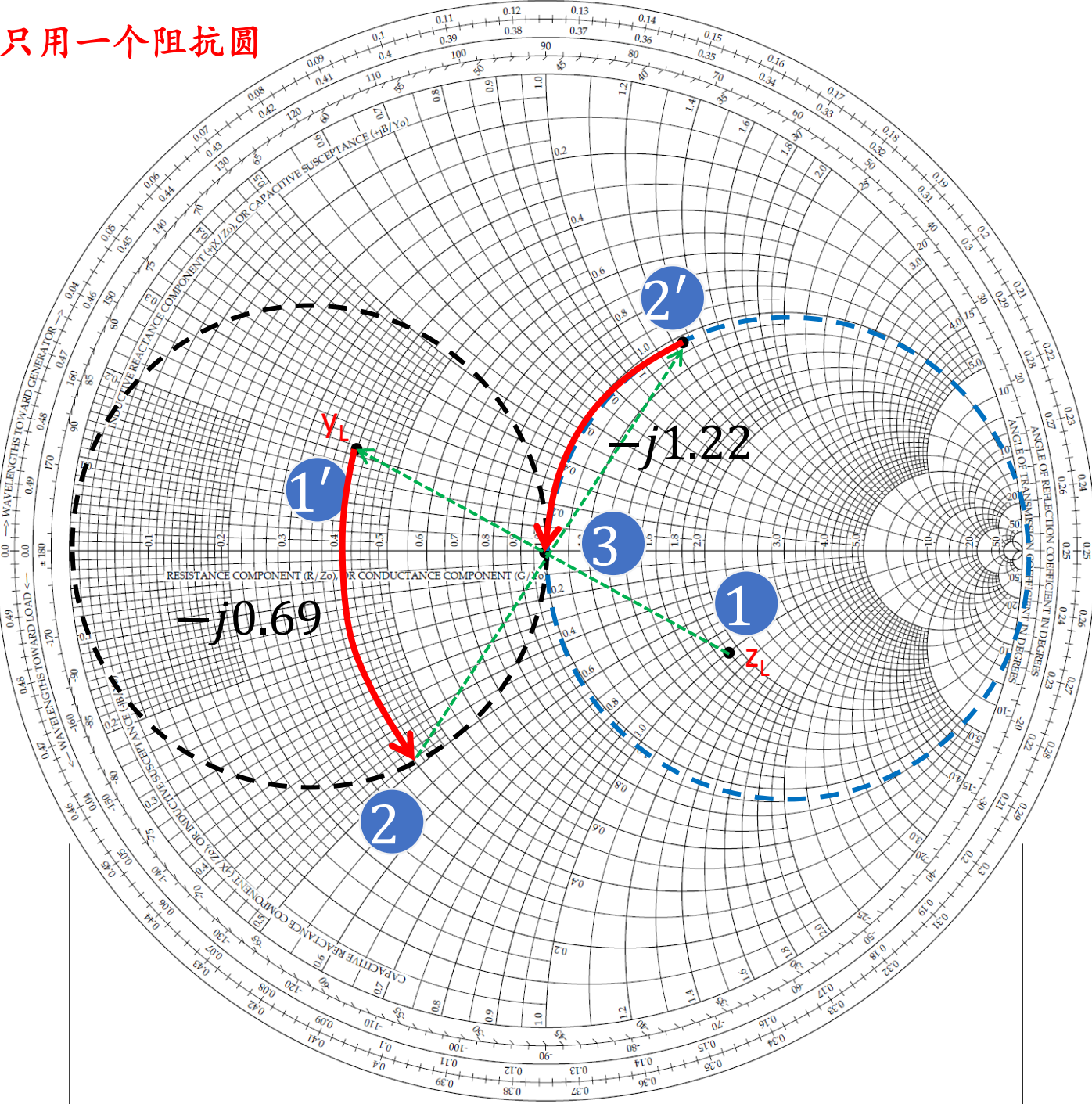
$$C = \frac{1}{1.22 \cdot 2\pi \cdot 500e6 \cdot 100} = 2.61 \text{ pF}$$



Solution 2

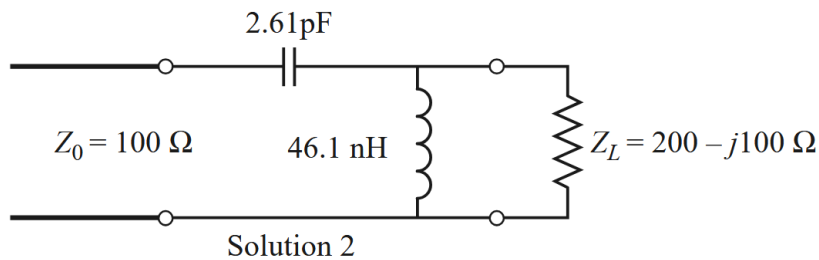


只用一个阻抗圆



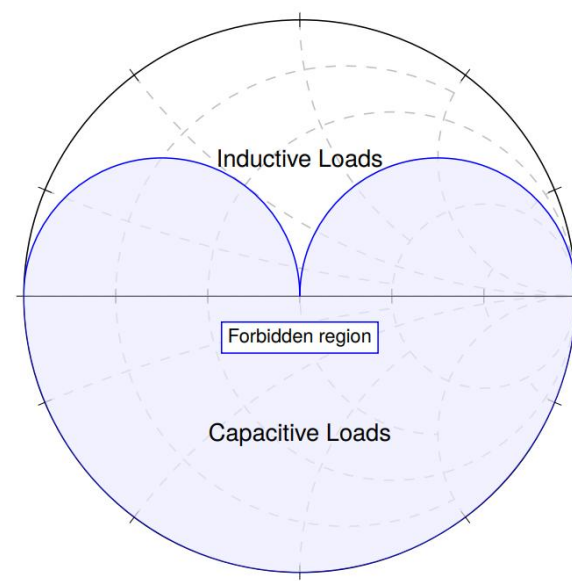
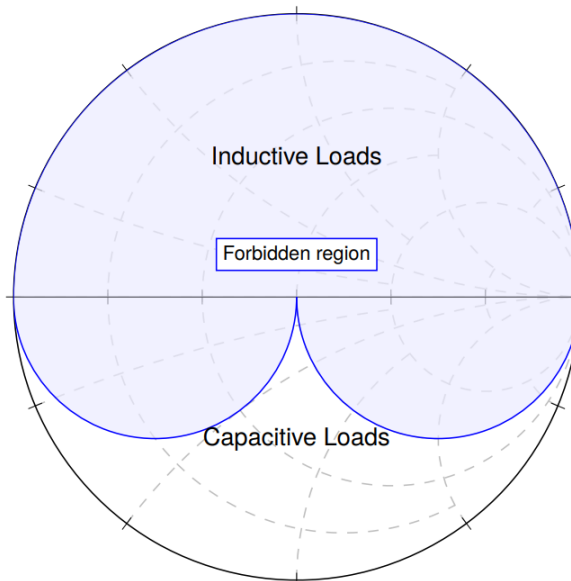
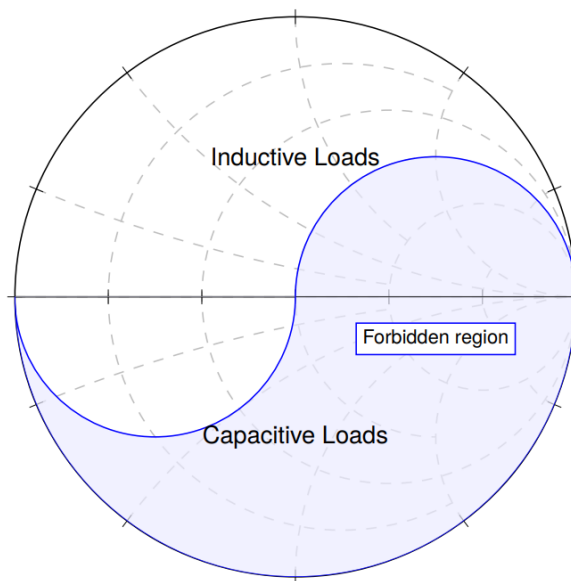
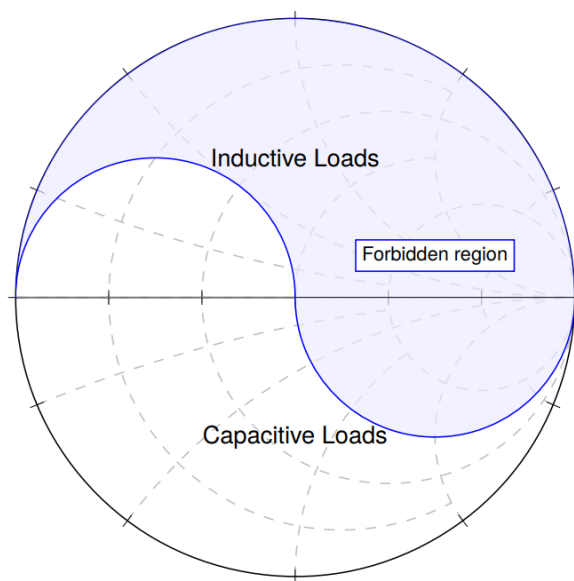
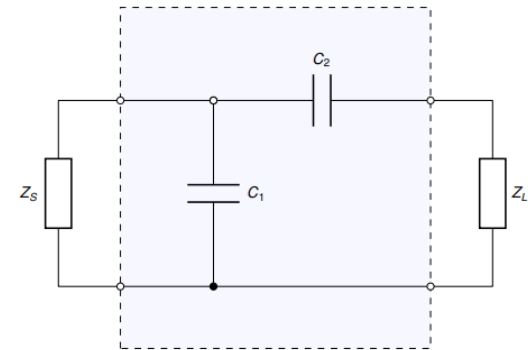
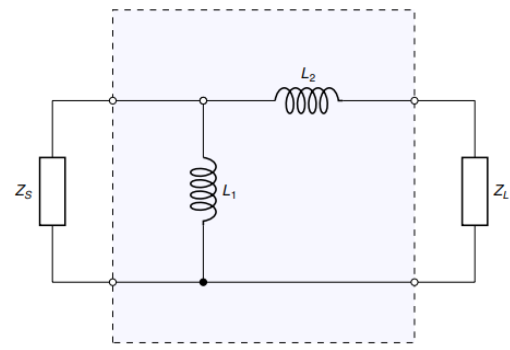
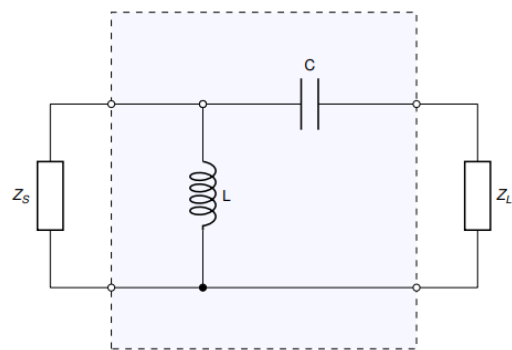
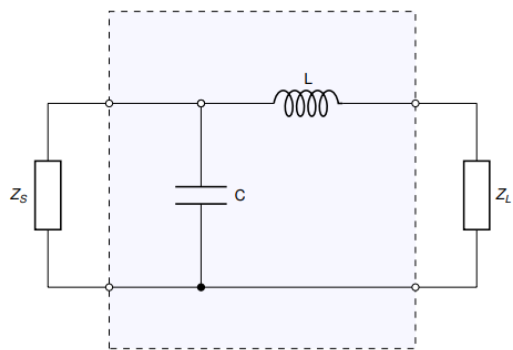
第一步：1' → 2: 导纳-j0.69
并联电感： $-(jZ_0)/\omega L = -j0.69$
$$L = \frac{100}{0.69 \cdot 2\pi \cdot 500e6} = 46.1 \text{ nH}$$

第二步：2' → 3: 阻抗-j1.22
串联电容： $-j1/(\omega CZ_0) = -j1.22$
$$C = \frac{1}{1.22 \cdot 2\pi \cdot 500e6 \cdot 100} = 2.61 \text{ pF}$$



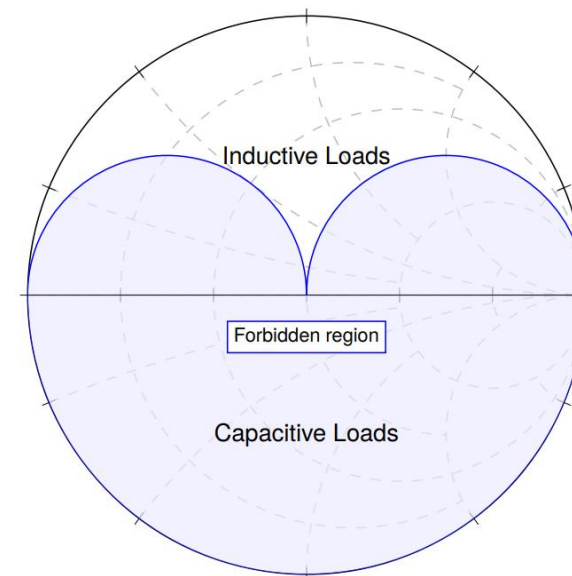
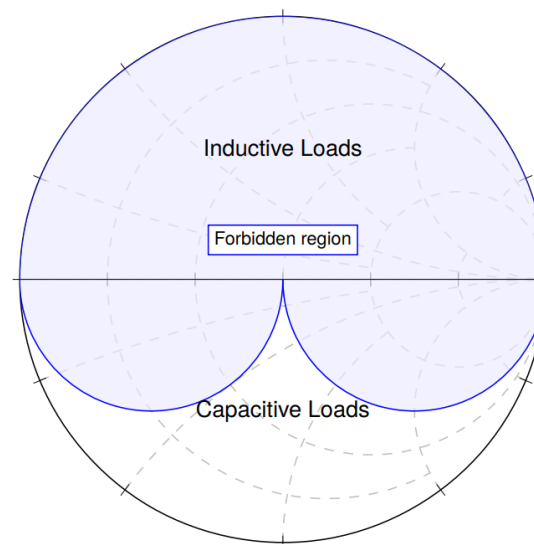
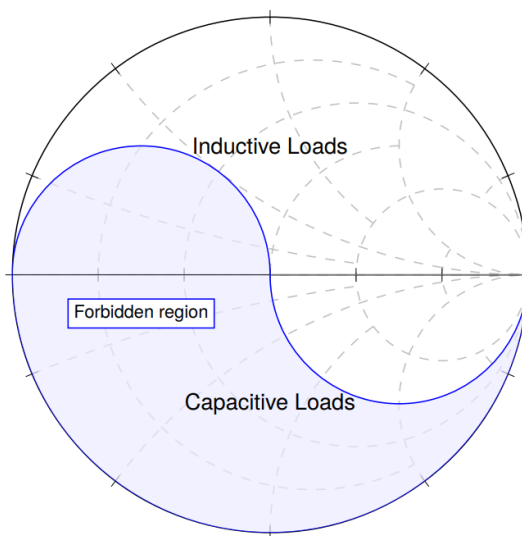
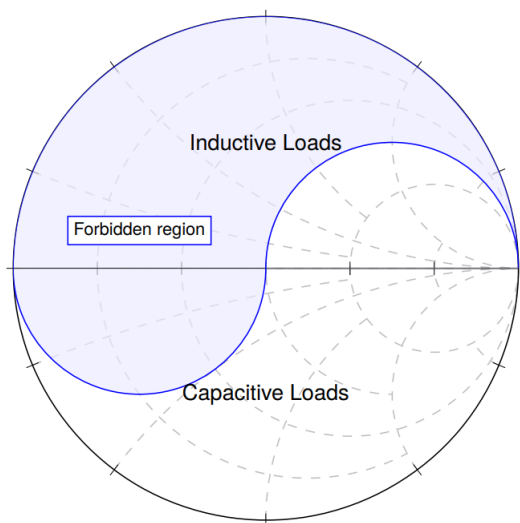
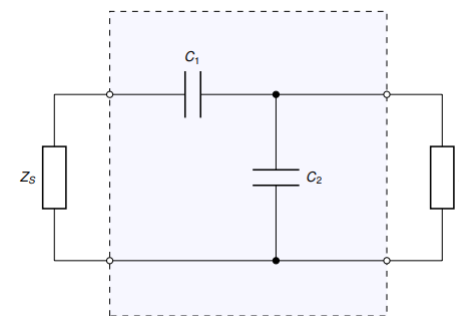
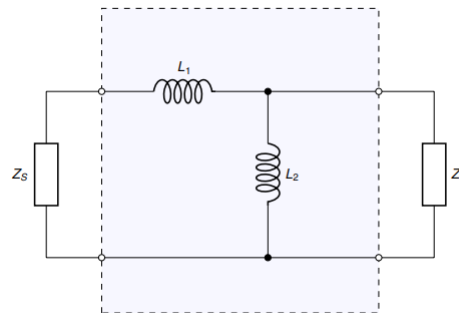
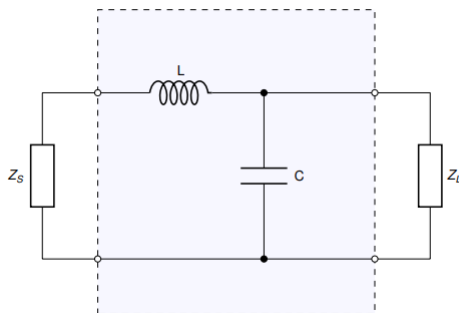
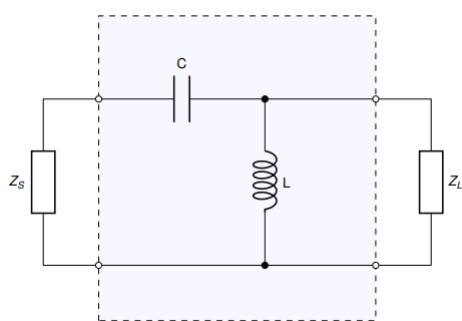
集总元件—L型匹配的匹配禁区

L型匹配电路有8种不同结构，每一种匹配电路都存在难以实现匹配的负载，在史密斯圆图上是一个区域，称为**匹配禁区**。



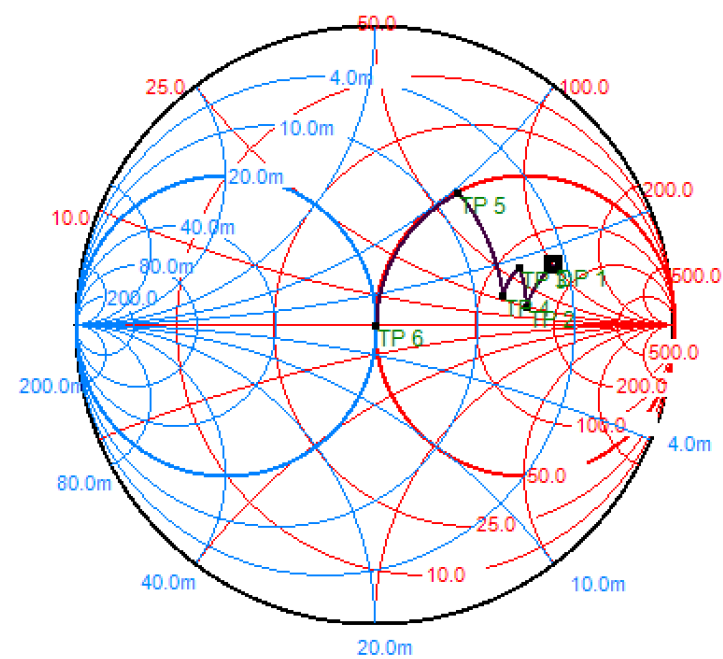
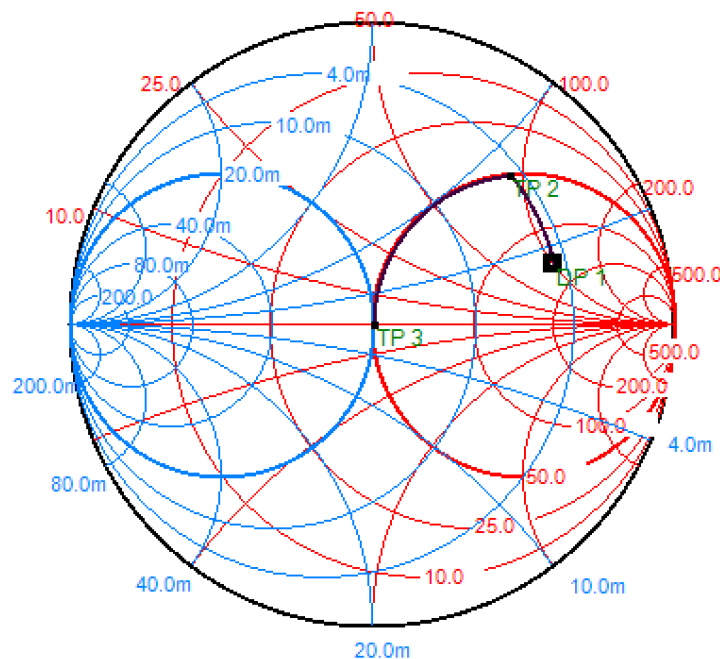
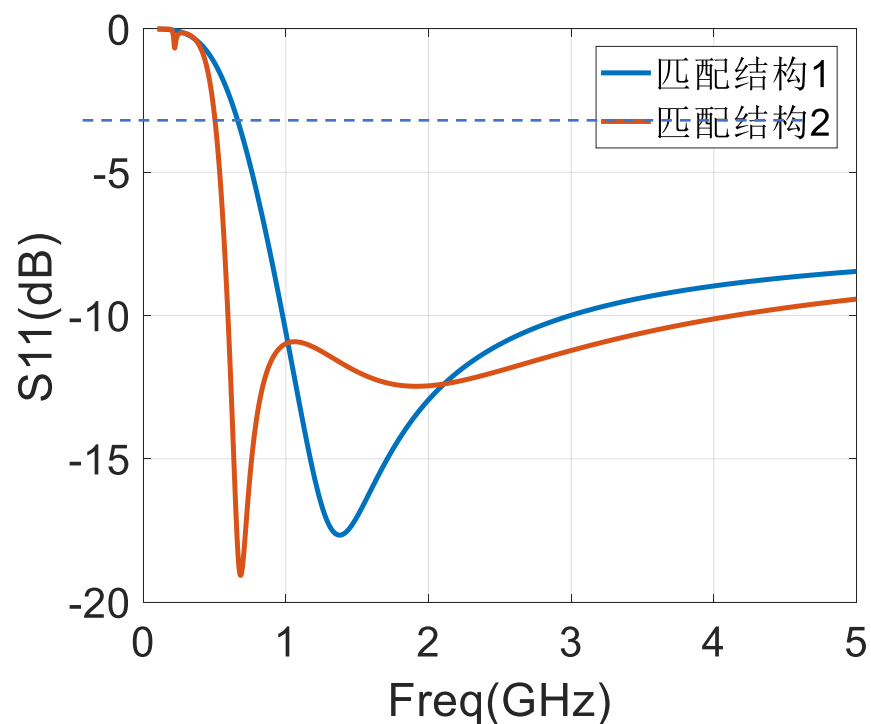
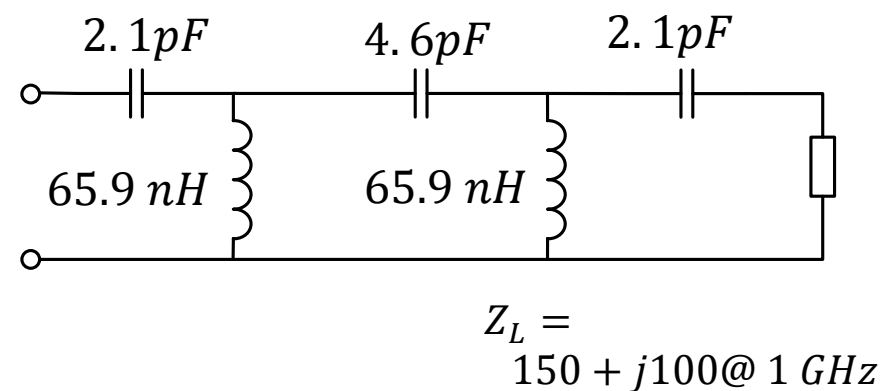
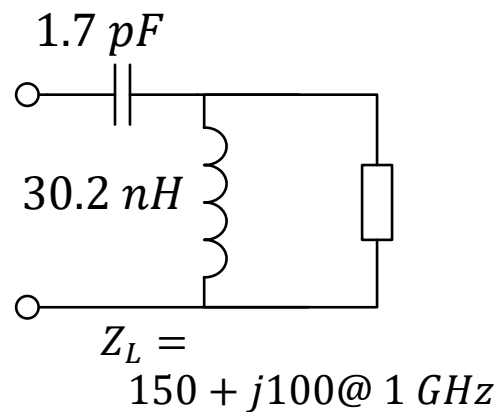
集总元件—L型匹配的匹配禁区

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集总元件阻抗匹配-Q值等值线

- ✓ L型匹配的问题在于带宽窄, 且固定
- ✓ 宽带放大器需要宽带宽
- ✓ 振荡器需要极窄带宽



集总元件阻抗匹配

- 匹配网络可以用有载 Q_L 值描述：

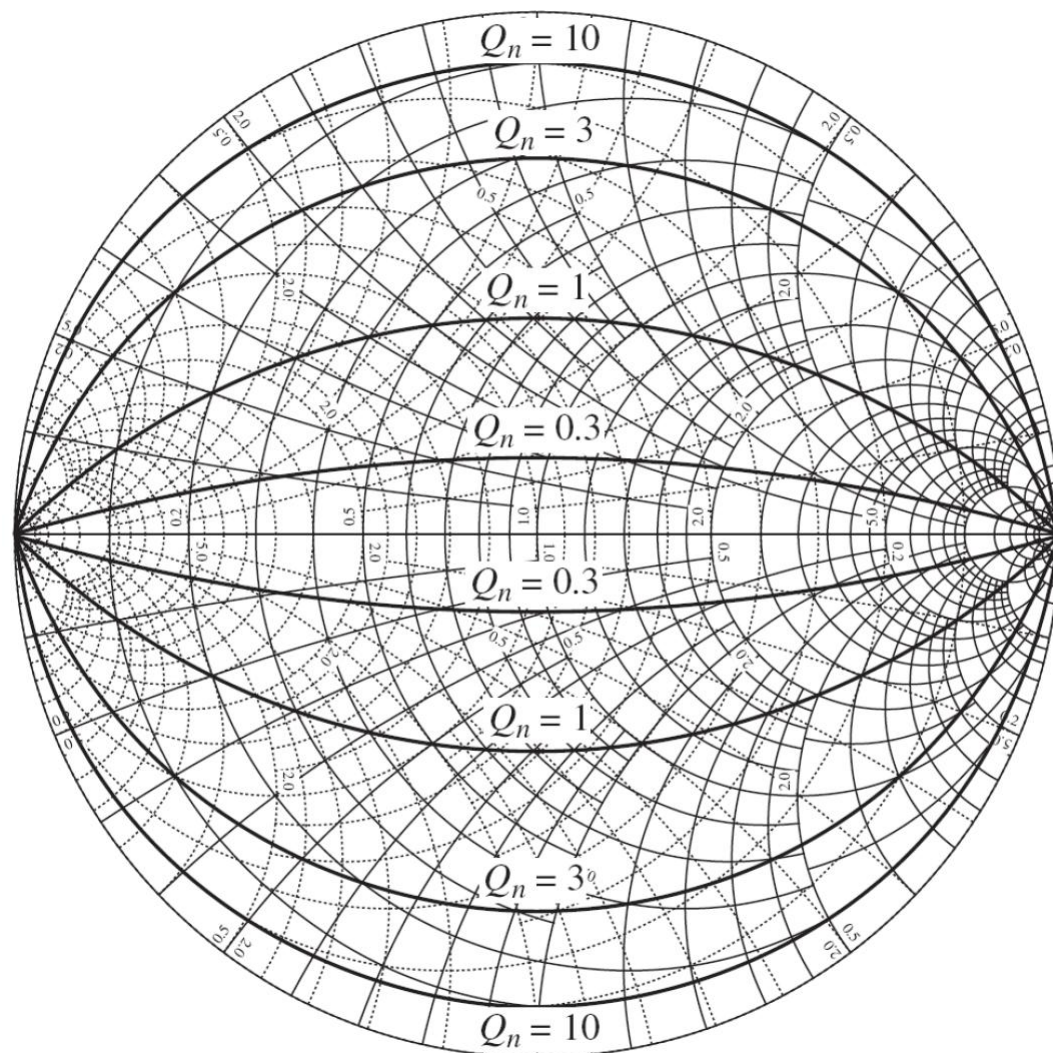
$$Q_L = \frac{f_0}{BW}$$

- Q_L 值描述复杂
- 使用节点 Q 值, Q_n
- 对于任意一个节点, 其值为 $Z_S =$

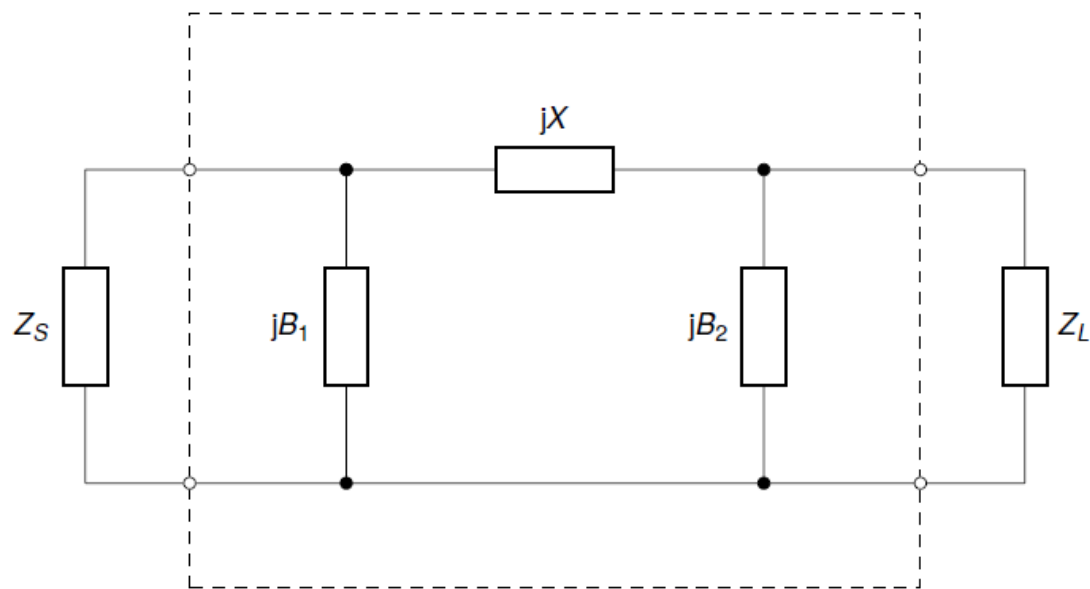
$$R_S + jX_S, \text{ 定义: } Q_n = \frac{|X_S|}{R_S}$$

$$z = r + jx = \frac{1 - \Gamma_r^2 - \Gamma_i^2}{(1 - \Gamma_r)^2 + \Gamma_i^2} + j \frac{2\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2}$$

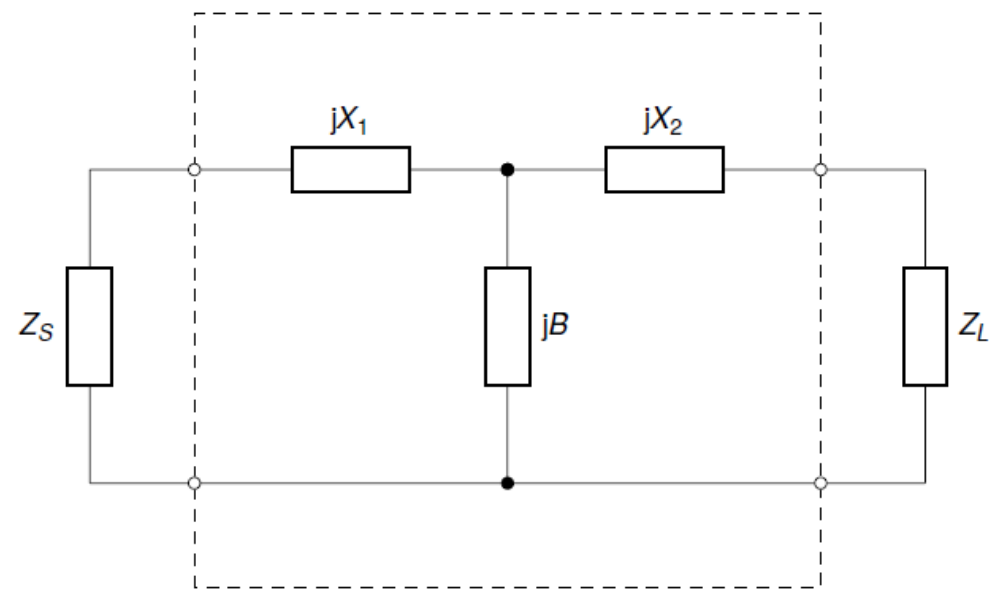
$$\Gamma_i^2 + \left(\Gamma_r \pm \frac{1}{Q_n} \right)^2 = 1 + \frac{1}{Q_n^2}$$



集总元件阻抗匹配-T型/PI型匹配网络

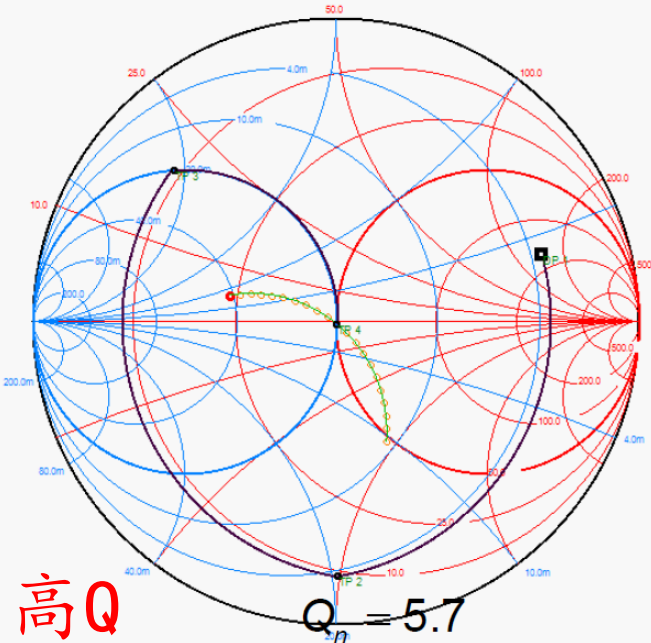
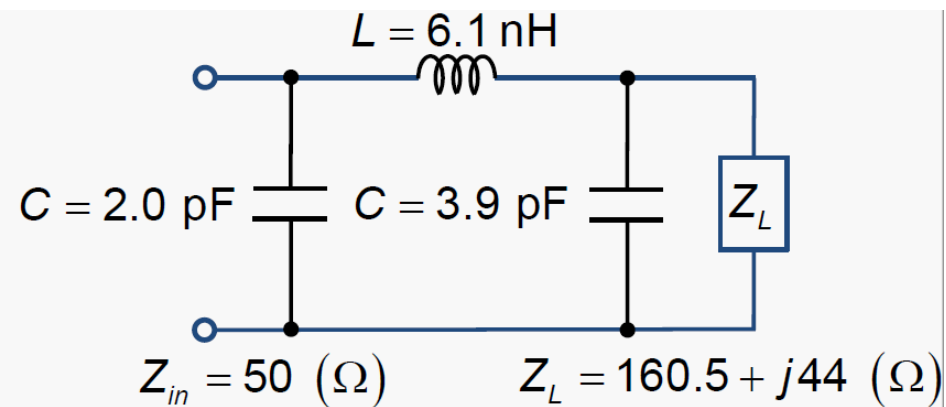
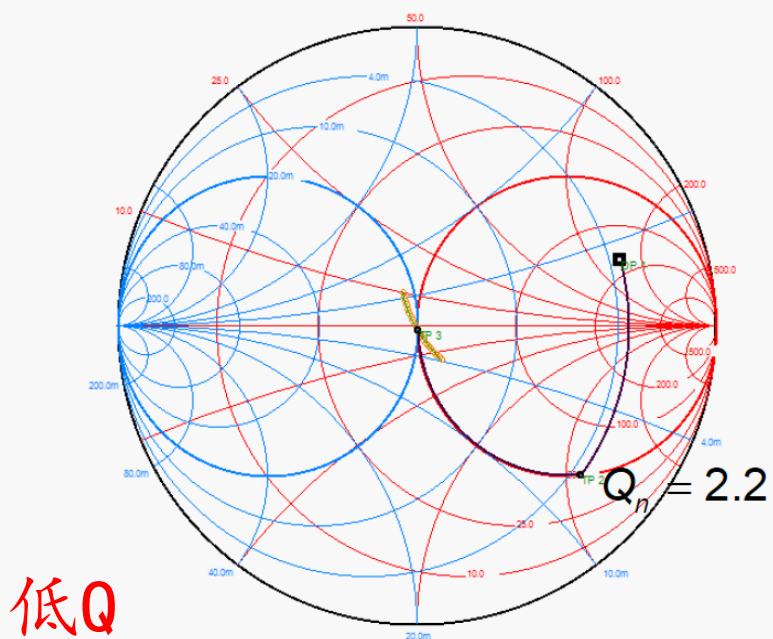
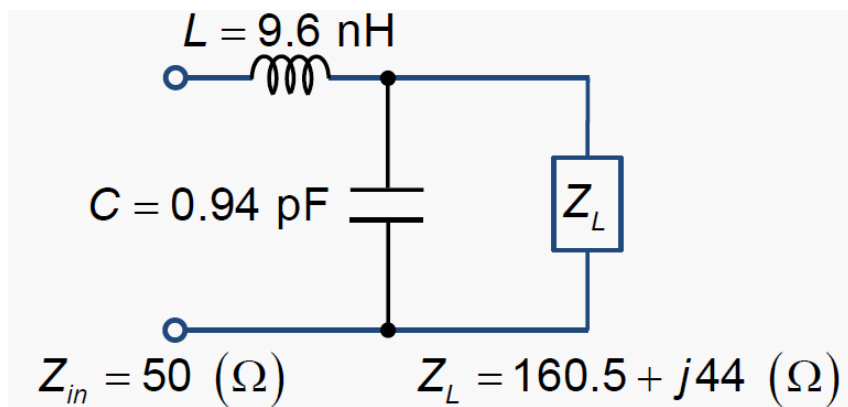


PI-型匹配网络



T-型匹配网络

集总元件阻抗匹配



基于传输线阻抗匹配-单枝节匹配

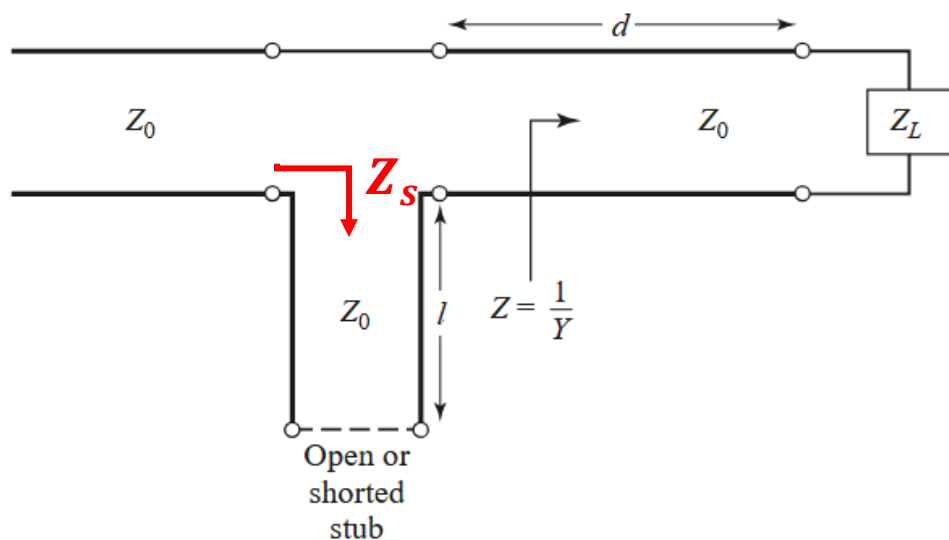
- LC集总元件具有自谐振频率
- 微波LC元件的损耗较大
- LC的元件值固定



无损传输线匹配

$$Z_s = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l} = jZ_0 \tan \beta l \quad (\text{Short})$$

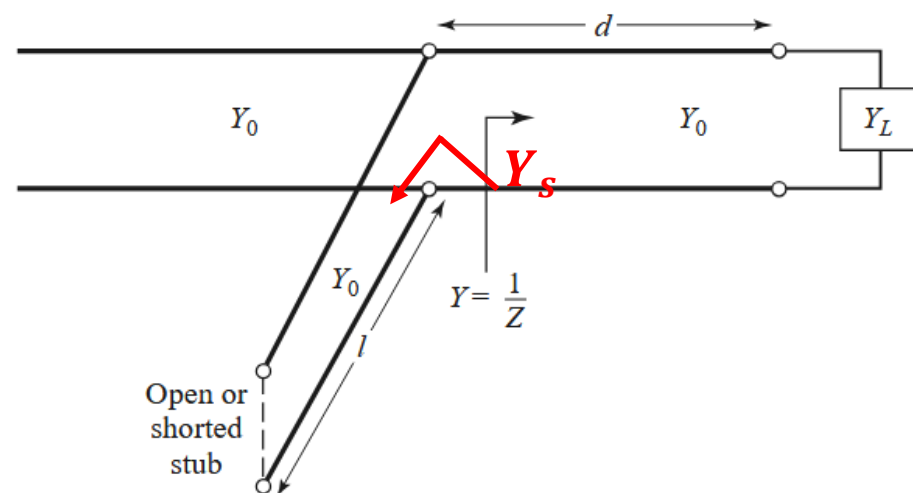
$$= \frac{Z_0}{j \tan \beta l} \quad (\text{Open})$$



串联开/短路枝节匹配

$$Y_s = \frac{1}{jZ_0 \tan \beta l} \quad (\text{Short})$$

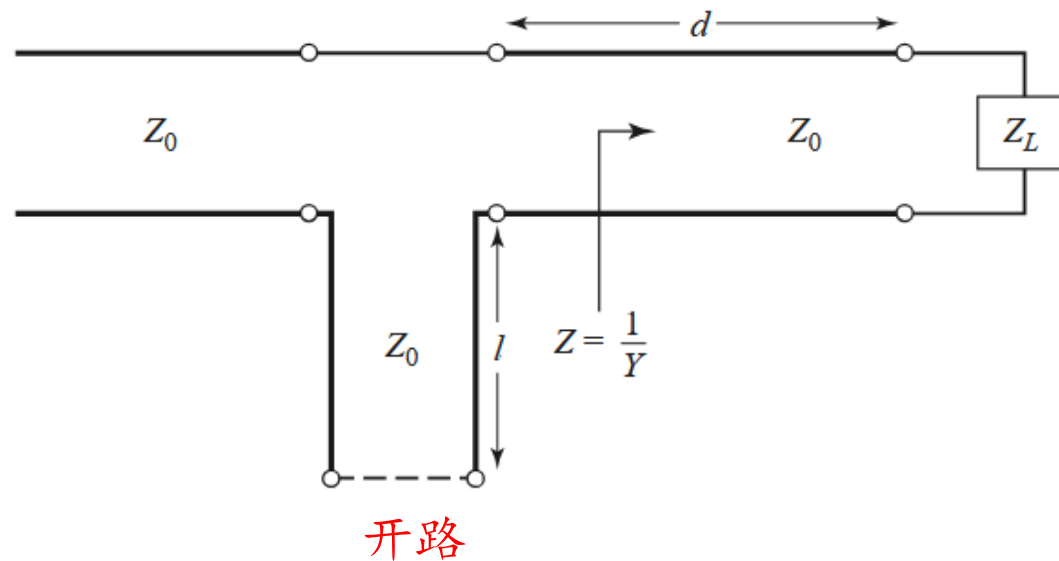
$$= \frac{j \tan \beta l}{Z_0} \quad (\text{Open})$$



并联开/短路枝节匹配

基于传输线阻抗匹配-串联单枝节匹配

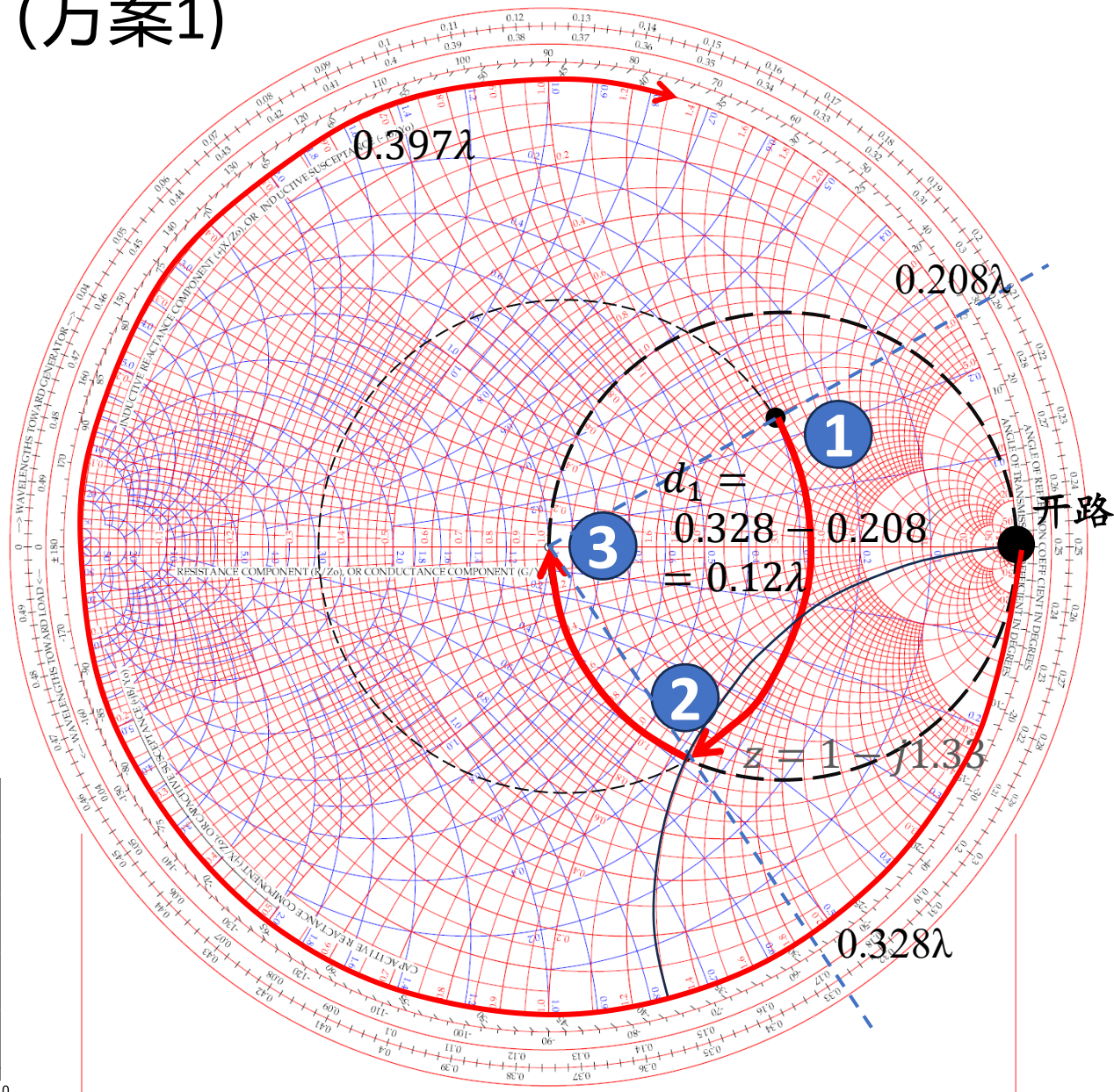
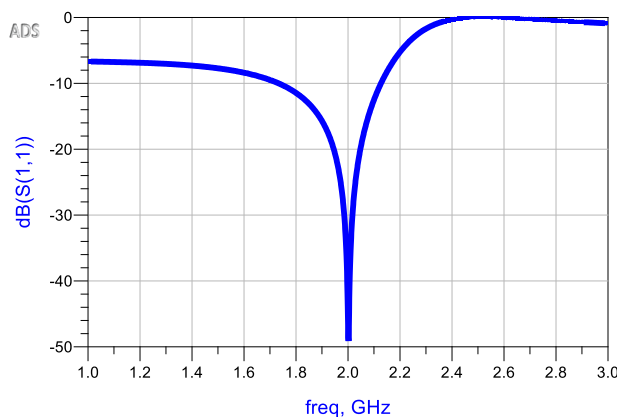
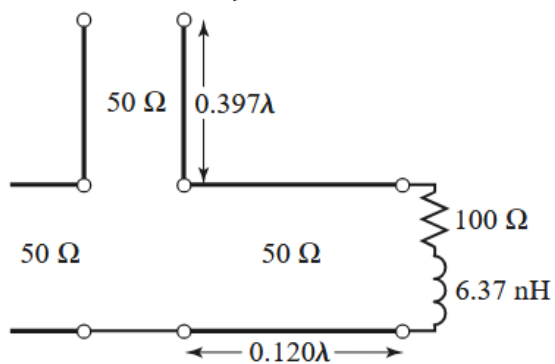
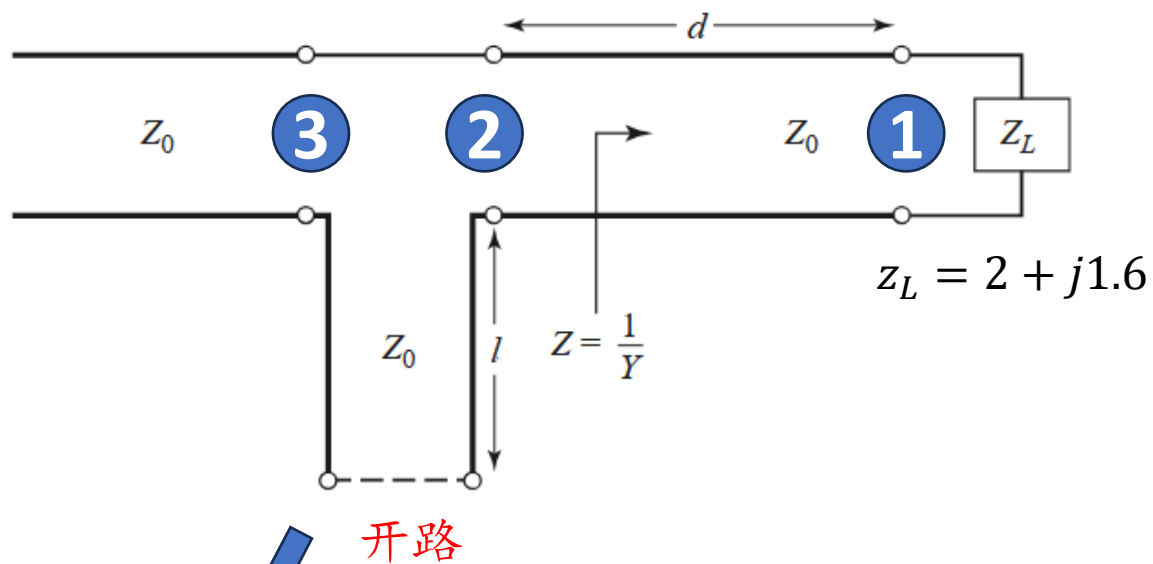
【例题】对于负载阻抗 $Z_L = 100 + j80$ ，设计串联单枝节（开路）网络，将该负载匹配至 $50\ \Omega$ 传输线。假设负载在 $2\ \text{GHz}$ 处匹配，且负载由电阻与电容串联组成，绘制每种解决方案在 $1\ \text{GHz}$ 至 $3\ \text{GHz}$ 频率范围内的反射系数幅度图。



基于传输线阻抗匹配-串联单枝节匹配 (方案1)

$$z = 1 - j1.33$$

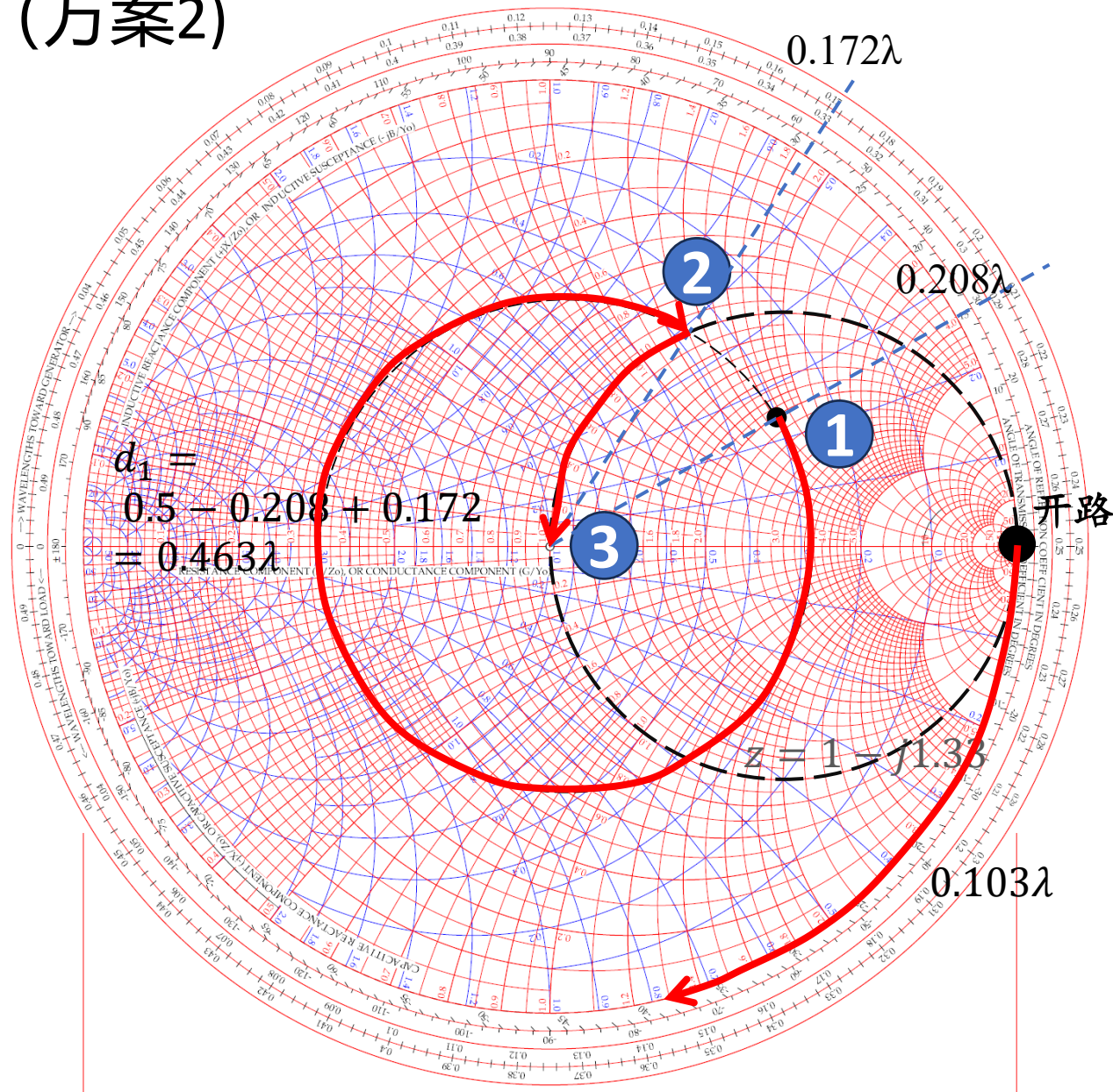
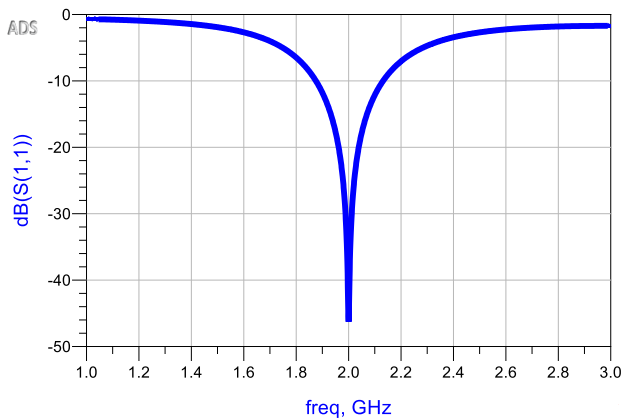
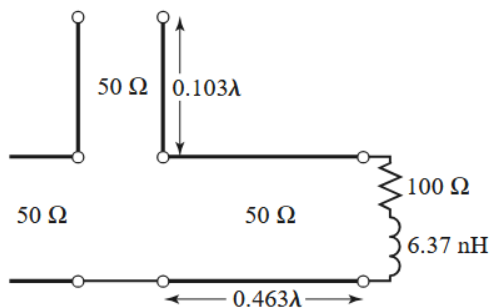
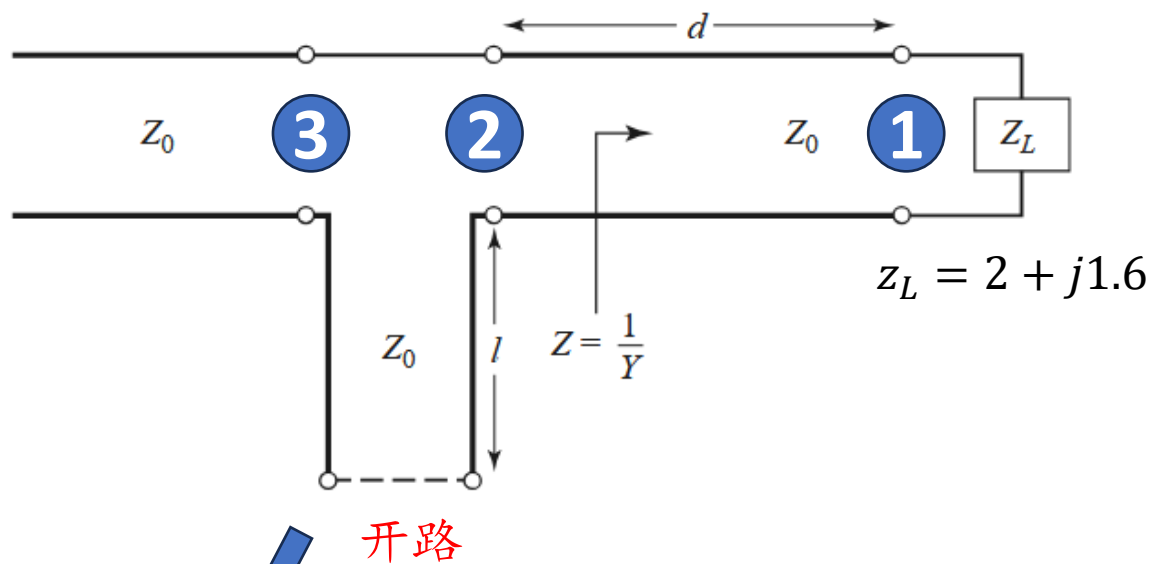
$$\Gamma(l) = \Gamma_L e^{-2j\beta l}$$



基于传输线阻抗匹配-串联单枝节匹配 (方案2)

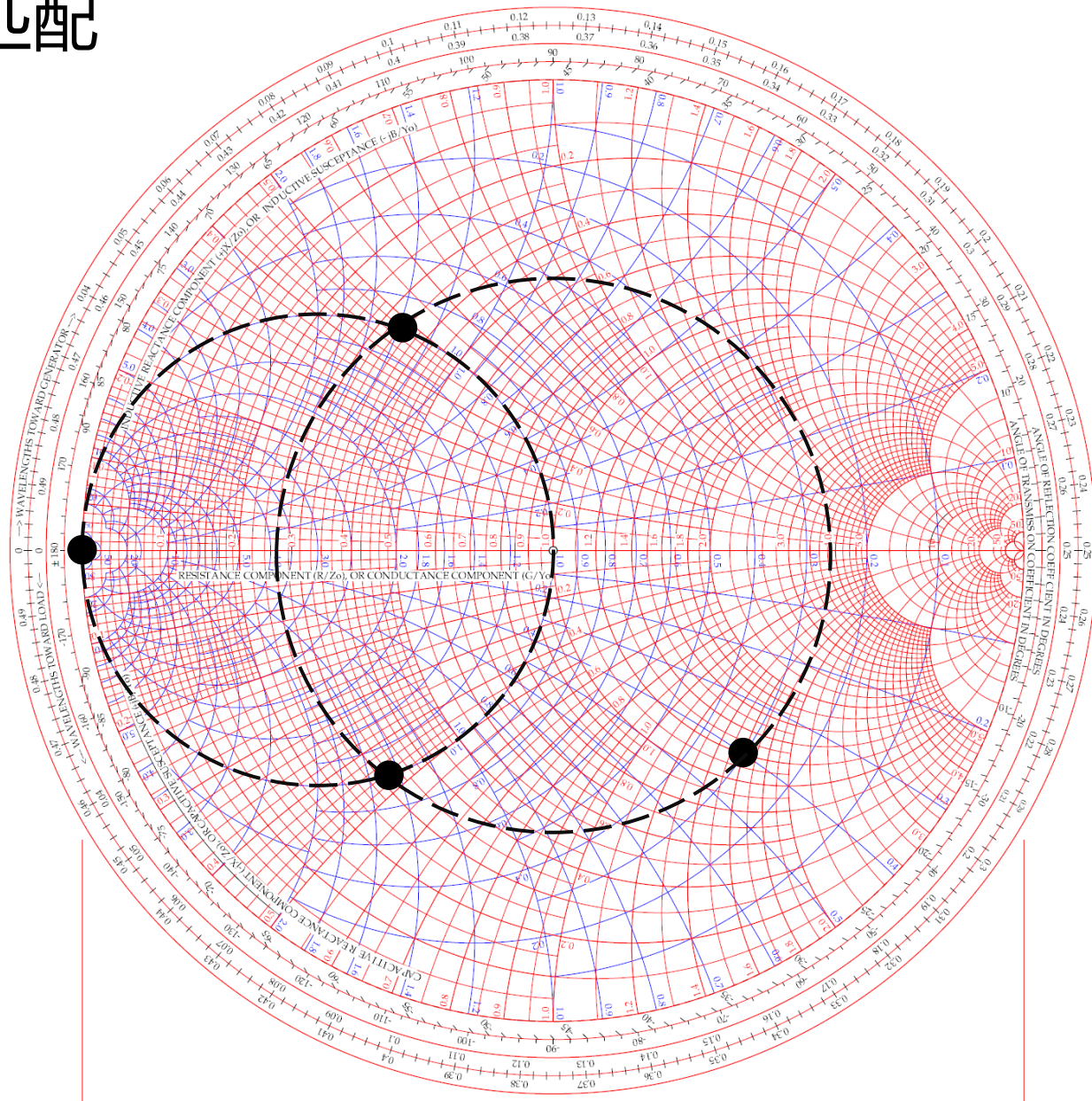
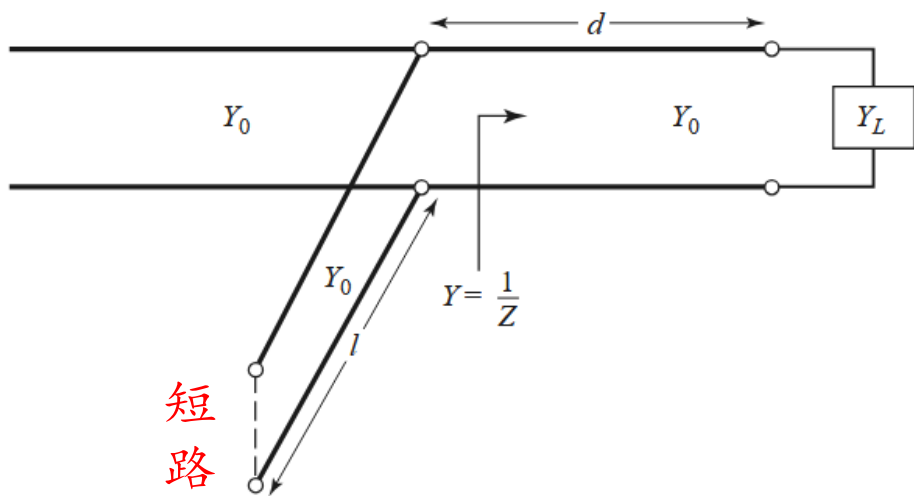
$$z = 1 + j1.33$$

$$\Gamma(l) = \Gamma_L e^{-2j\beta l}$$



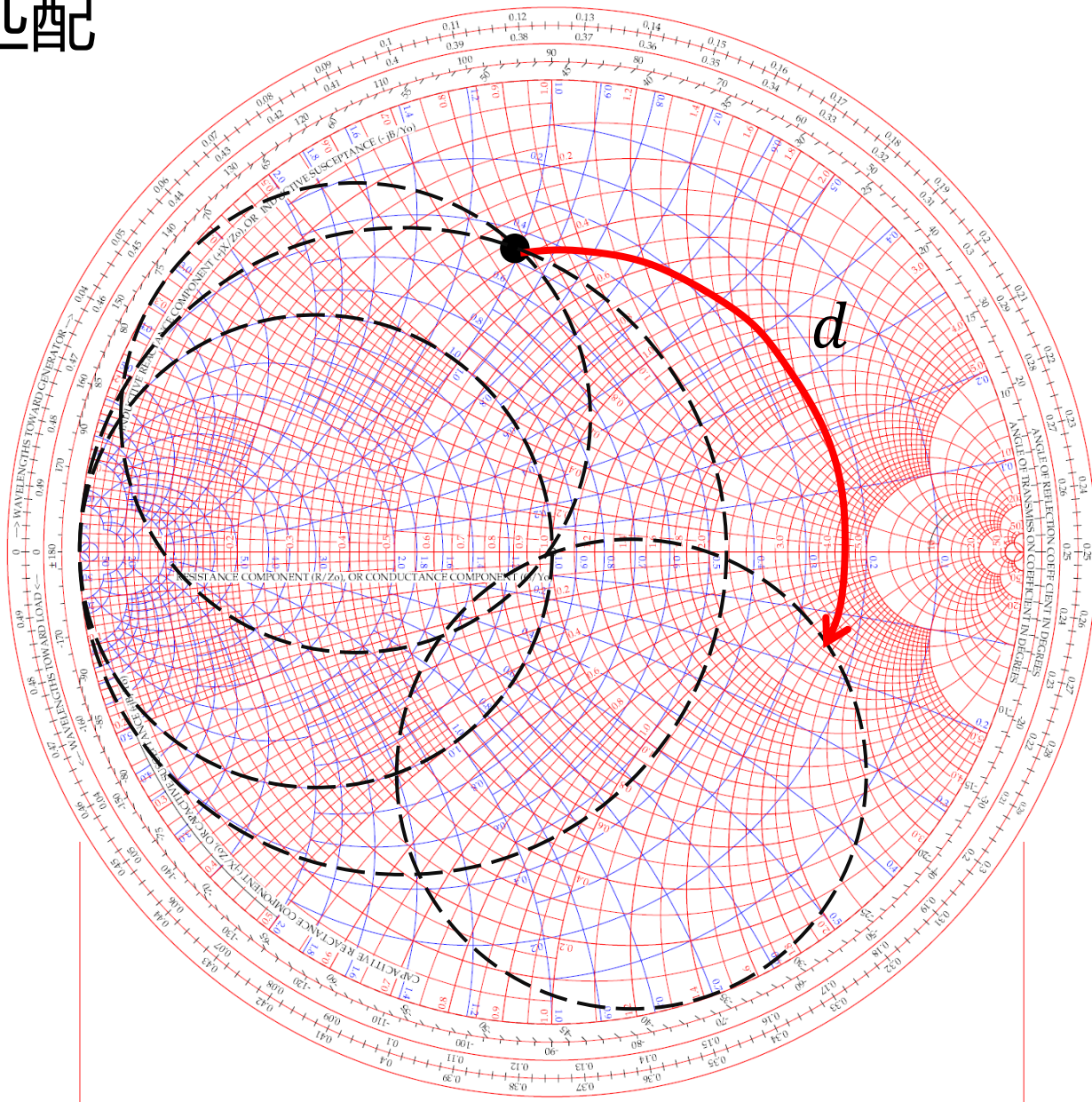
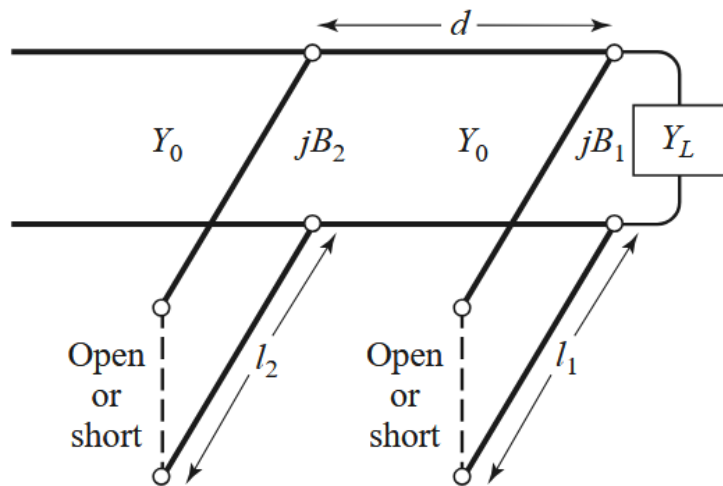
基于传输线阻抗匹配-并联单枝节匹配

【作业】对于负载阻抗 $Z_L = 60 - j80$ ，设计并联单枝节（短路）网络，将该负载匹配至 50Ω 传输线。假设负载在 2 GHz 处匹配，且负载由电阻与电容串联组成，绘制每种解决方案在 1 GHz 至 3 GHz 频率范围内的反射系数幅度图。



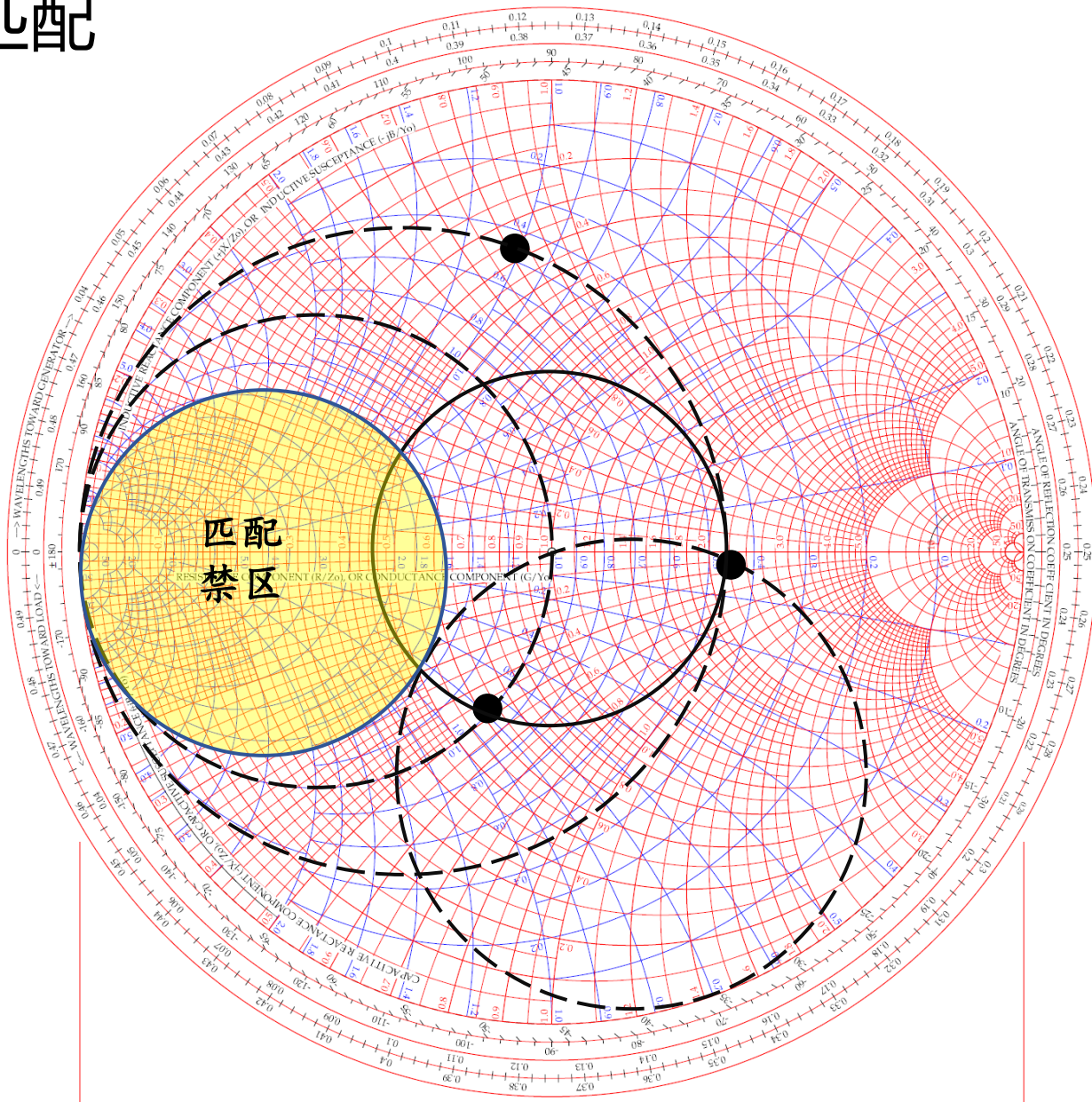
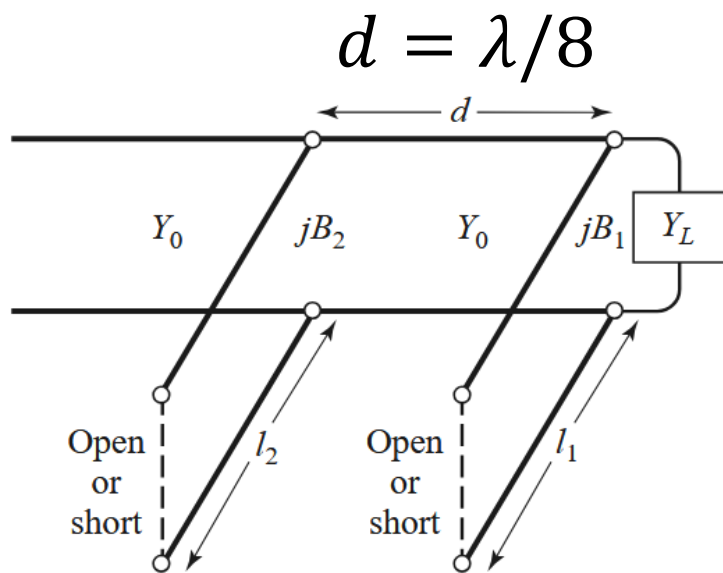
基于传输线阻抗匹配-并联双枝节匹配

- 单枝节匹配的缺点在于长度 d 是随着负载变化的
- 有没有一种方式，传输线的长度不变，而仅需改变枝节的长度即可匹配变化的负载？



基于传输线阻抗匹配-并联双枝节匹配

- 单枝节匹配的缺点在于长度 d 是随着负载变化的
- 有没有一种方式，传输线的长度不变，而仅需改变枝节的长度即可匹配变化的负载？



多枝节宽带匹配

